CME

Quality Indicators for EUS

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Endoscopic innovations remain at the forefront of the most exciting advances in gastroenterology and hepatology. This is best exemplified by endoscopic ultrasound (EUS) and specifically interventional EUS, which has grown exponentially in the last decade. EUS was first developed in the 1980s for staging of upper gastrointestinal (GI) malignancies. Indications then steadily grew to include staging of cancers throughout the digestive system, characterization of subepithelial lesions (SELs), and assessment of pancreaticobiliary disorders. Diagnostic capabilities expanded in the 1990s when tissue acquisition with EUS-guided fine needle aspiration (FNA; EUS-FNA) became available. This ability to advance a needle into a target structure under sonographic visualization led to the development of procedures like pseudocyst drainage and fine-needle injection (1). On this background, the field of interventional EUS has recently taken off, allowing "bridges" to be built between anatomic structures within and outside the GI tract for myriad indications such as drainage, bypass, and altered-anatomy endoscopic retrograde cholangiopancreatography (ERCP). However, the excitement generated by these innovations must be tempered by a commitment to high-quality practice characterized by meticulous training and responsible diffusion aiming to maximize benefit while minimizing the potential for overuse and adverse events (AEs).

The American College of Gastroenterology (ACG) and the American Society for Gastrointestinal Endoscopy (ASGE), the 2 co-sponsors of this series, remain maximally committed to delivering high-quality endoscopic care. This commitment is reflected, in part, by quality indicator documents that aim to provide endoscopists and practices a framework through which quality improvement efforts can be operationalized. This document presents the quality indicators for EUS (Table 1).

Generally speaking, quality in health care is the degree to which health services for individuals and populations increase the likelihood of a desired outcome and are consistent with current professional knowledge (2). In the endoscopic context, a high-quality procedure is clearly indicated during which relevant diagnoses are established or excluded, any therapy provided is appropriate and effective, and harm is minimized to the greatest extent possible. Toward this ideal, quality indicators have been developed to compare the performance of an individual or a group of individuals to available standards. Quality indicators can be divided into 3 categories: structural measures, assessing characteristics of the entire healthcare environment (eg, availability and maintenance of endoscopy equipment at a hospital); process measures, assessing performance during the delivery of care (eg, proportion of patients who undergo biopsy sampling when Barrett's esophagus is suspected); and outcome measures, assessing the results of the care that is provided (eg, proportion of patients who survive at least 30 days after endoscopic therapy for a bleeding gastric ulcer). By developing evidence-based quality measures, establishing performance targets—known as benchmarks—that reflect high-quality practice, measuring performance against these benchmarks, and implementing interventions to improve this performance, the quality of care delivered at the endoscopist, practice, and field levels can be maximized.

METHODOLOGY

This work represents the third iteration of the ACG-ASGE quality indicators document pertaining to EUS. The first version of this document was published by the ACG-ASGE Task Force on quality indicators for endoscopic ultrasonography in 2006 (3) and was revised in 2015 (4). This revision integrates new data relevant to existing quality indicators and introduces new indicators as appropriate based on interval progress in the field. Although this document focuses on EUS, the indicators that are common to all GI endoscopic procedures are presented in detail (5). These common indicators are addressed herein only insofar as the discussion needs to be modified specifically to relate to EUS.

As in preceding versions, we prioritized indicators with wide-ranging clinical implications associated with variation in practice and outcomes and validated in clinical studies. When supportive data were absent, indicators of particular clinical importance were chosen by expert consensus. Substantial progress has been made in the last decade in our ability to measure performance. However, feasibility and efficiency challenges still need to be addressed. Nevertheless, the task force elected to include a limited number of highly relevant but not yet easily measurable indicators to promote their eventual adoption.

As in preceding versions, quality indicators are divided into 3 time periods: preprocedure, intraprocedure, and post-procedure. Additionally, each quality indicator is classified as an

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Table 1. **Quality indicators for EUS procedures with associated performance targets**

Quality indicator	Strength of recommendation	Measure type	Performance target (%)
Preprocedure			
1. Frequency with which EUS is performed for an indication that is included in a published standard list of appropriate indications, and the indication is documented (priority indicator)	1C+	Process	>90
2. Frequency with which informed consent is obtained, including a specific discussion of the risks associated with EUS, and the consent is documented	3	Process	>98
3. Frequency with which prophylactic antibiotics are administered for appropriate indications	1B	Process	>98
4. Frequency with which EUS is performed or supervised by an endosonographer who is fully trained and appropriately credentialed to perform that particular procedure	3	Process	>98
Intraprocedure			
5. Frequency with which the visualization of relevant structures, specific to the indication for the EUS, is documented	3	Process	>98
6. Frequency with which luminal GI cancers are staged using the American Joint Committee on Cancer and Union for International Cancer Control TNM staging system	3	Process	>98
7. The frequency with which the echogenicity and wall layer of origin of subepithelial masses are documented (priority indicator)	3	Process	>98
8. The frequency with which a diagnostically adequate EUS-guided liver biopsy sample is obtained	1B	Outcome	>85
9. The frequency with which a pancreatic mass ≥10 mm is identified with EUS	2C	Outcome	≥90
10. The frequency with which pancreatic mass measurements, vascular and lymph node involvement, liver metastases, and ascites are evaluated and documented	3	Process	≥90
11. The frequency with which a diagnostic specimen is obtained by EUS-guided sampling of a malignant pancreatic mass (priority indicator)	1C+	Outcome	≥87
12. Frequency with which technical success in EUS-guided pancreatic fluid collection drainage is achieved (priority indicator)	1B	Outcome	≥92
13. Frequency with which technical success in EUS-guided gallbladder drainage is achieved	2B	Outcome	>90
14. Frequency with which technical success in EUS-guided biliary drainage is achieved	1B	Outcome	>85
15. Frequency with which technical success in EUS-guided gastroenterostomy is achieved	2C	Outcome	>85
16. Frequency with which technical success in EUS-directed transgastric ERCP is achieved	1C	Outcome	>92

Table 1. (*continued*)

Quality indicator	Strength of recommendation	Measure type	Performance target (%)
Postprocedure			
17. Frequency with which a complete report is created	3	Process	>98
18. Frequency with which the incidence of AEs after diagnostic and interventional EUS are documented (priority indicator)	3	Process	>98
19. Frequency with which AEs after diagnostic EUS, including EUS-FNA/FNB/FNI\liver biopsy sampling, occur	1C+	Outcome	Perforation <0.5 Infection <1 Acute pancreatitis <1 Clinically significant bleeding <1 Clinically significant bleeding <5 after EUS-guided liver biopsy sampling
20. Frequency with which AEs related to interventional EUS procedures occur	2C	Outcome	Overall AEs of <10 for EUS-guided pancreatic fluid collection drainage <20 for EUS-guided gallbladder drainage <25 for EUS-guided biliary drainage <15 for EUS-guided gastroenterostomy and EUS-directed transgastric ERCP

AEs, adverse events; ERCP, endoscopic retrograde cholangiopancreatography; EUS, endoscopic ultrasound; FNA, fine needle aspiration; FNB, fine needle biopsy; FNI, fine needle injection; GI, gastrointestinal.

outcome or process measure. Although outcome measures are generally considered more impactful toward improving quality of care, some can be difficult or impossible to measure in routine clinical practice because of the need for large amounts of data and/or long-term follow-up and because other factors may confound their measurement or interpretation. In such cases, process indicators are provided as surrogate measures of high-quality endoscopic practice. The relative value of a process indicator hinges on the evidence that supports its association with a clinically relevant outcome, and such process measures were emphasized. The measures in this document pertain directly to endoscopic care. Of course, the quality of care delivered to patients is influenced by additional factors, including those related to the facilities where endoscopy is performed. These structural measures are covered in a separate document dedicated to unit-level quality (6).

For this revision, the task force critically appraised existing quality indicators and determined whether to maintain or remove them by consensus based on several factors, including ongoing relevance and strength of evidence. Additionally, task force members proposed new quality indicators and, if appropriate, were adopted by consensus based on similar considerations. For each indicator, the authors identified relevant articles through a systematic search of PubMed (US National Library of Medicine, National Institutes of Health) from January 2014 (the date of the last update of this document) through March 2024. The search strategies for each indicator included a combination of subject headings (MeSH in PubMed) and pertinent key words. English language restrictions were applied. To identify additional articles, the authors reviewed PubMed's "similar articles" and manually searched reference lists of relevant articles. Search strategies for the selected indicators were facilitated by health science librarians with expertise in systematic review. Searches that were judged to be incomplete after librarian assessment were repeated. Based on this revised literature review, the strength of evidence for each indicator was evaluated according to a previously used framework (Table 2). Within this framework, the strength of each quality indicator is divided across a spectrum from "1A," denoting a strong quality indicator that can be applied to most clinical settings, to "3," denoting a weak quality indicator because of the absence of evidence requiring reliance on expert opinion only. The strength of the evidence grade for each indicator was established by author consensus.

The process and outcome measures included in this document are attached to a performance target, and therefore each measure is considered a quality indicator. The task force selected performance targets based on published benchmarking data informed by literature review. In the absence of available data, when expert consensus considered failure to perform a given quality indicator a "never event," such as failure to monitor vital signs during sedation, the performance target was expressed as >98% because only in exceptional circumstances would the quality indicator not be fulfilled.

It is important to emphasize that the included quality indicators and associated performance targets do not necessarily reflect the standard of care, credentialing requirements, or training standards and using any of the quality indicators in this document as such is a misuse. Instead, the quality indicators are designed to serve as a framework for quality improvement efforts. To guide continual improvement for endoscopists and units in varying phases of their quality journey, the task force again recognized a subset of indicators deemed "priority indicators," based on their clinical relevance and importance, evidence that performance varies significantly in clinical practice, and feasibility of measurement (a function of the number of procedures needed to obtain an accurate measurement with narrow confidence

Table 2. Strength of recommendations

Strength of recommendation	Clarity of benefit	Methodologic strength supporting evidence	Implications
1A	Clear	Randomized trials without important limitations	Strong recommendation; can be applied to most clinical settings
1B	Clear	Randomized trials with important limitations (inconsistent results, nonfatal methodologic flaws)	Strong recommendation; likely to apply to most practice settings
1C+	Clear	Overwhelming evidence from observational studies	Strong recommendation; can apply to most practice settings in most situations
1C	Clear	Observational studies	Intermediate-strength recommendation; may change when stronger evidence is available
2A	Unclear	Randomized trials without important limitations	Intermediate-strength recommendation; best action may differ depending on circumstances or patient or societal values
2B	Unclear	Randomized trials with important limitations (inconsistent results, nonfatal methodologic flaws)	Weak recommendation; alternative approaches may be better under some circumstances
2C	Unclear	Observational studies	Very weak recommendation; alternative approaches are likely to be better under some circumstances
3	Unclear	Expert opinion only	Weak recommendation; likely to change as data become available

Adapted from Guyatt G, Sinclair J, Cook D, et al. Moving from evidence to action. Grading recommendations: A qualitative approach. In: Guyatt G, Rennie D, editors. Users' Guides to the Medical Literature. AMA Press: Chicago, IL, 2002:599–608.

intervals (CIs) and the ease of measurement). We believe that quality improvement efforts should initially focus on the priority indicators and then progress to include other indicators once it is ascertained that endoscopists perform above recommended thresholds, either at baseline or after corrective interventions.

PREPROCEDURE QUALITY INDICATORS

The preprocedure period begins at the time of first contact between the patient and members of the endoscopy team and ends at the time of administration of sedation. Quality indicators for EUS during this period include appropriate indication, informed consent, and antimicrobial management.

1. Frequency with which EUS is performed for an indication that is included in a published standard list of appropriate indications and the indication is documented (priority indicator)

Level of evidence: 1C+

Performance target: >90%

Type of measure: Process

Established and emerging indications for EUS are listed in Table 3 and are divided according to whether they are diagnostic or therapeutic (7). The ever-expanding field of interventional EUS, discussed in detail in this document, continues to generate new indications. Strikingly, therapeutic procedures now account for >50% of accepted indications for EUS; this represents a significant increase from the prior documents (3,4).

Discussion. The beauty of endoscopy is the inherent innovation that accompanies it. To allow for the unimagined, we recognize that established indications are a starting point, and we must be willing to incorporate nuanced adaptations or altogether new techniques, devices, and accessories that accompany EUS. As such, unlike other procedures for which the appropriate indication performance targets are >95%, the 90% target for EUS allows latitude to "think outside of the box" as long as the clinical benefit and safety profile justify the novel intervention and transparency is maximized.

The indications for diagnostic EUS have remained largely unchanged. As long as there is no evidence of distant metastasis, EUS remains the mainstay of initial staging in the upper GI tract, such as for esophageal and gastric cancers. Rectal cancers are best staged with a combination of endorectal ultrasound (US) and magnetic resonance imaging (MRI).

EUS also plays an important part in the primary staging of pancreatic neoplasms, although its role in assessing vascular involvement in pancreatic carcinoma has been somewhat tempered by advances in cross-sectional imaging. Accordingly, over the past decade, there has been a shift away from proclaiming the superiority of either EUS or cross-sectional imaging for staging pancreaticobiliary malignancies and specifically for vascular involvement or invasion. Instead, growing clinical evidence supports the complementary roles of these modalities (8–10). The role of EUS in pancreaticobiliary malignancies is discussed in more detail later in this document. EUS for pancreatic cancer screening in high-risk individuals represents a new diagnostic indication and has been the focus of society guideline documents (11,12).

The advent of several therapeutic EUS-guided interventions, which now encompass an expanding list of indications absent in the prior quality indicator document, highlights the precise rationale for not mandating >95% compliance with established indications, allowing the natural evolution of novel endosonographic interventions. However, when "outside the box" applications of EUS are contemplated and performed, it is essential that all relevant stakeholders are involved in clinical decision-making and that the patient is fully informed of the emerging nature of the procedure.

Therapeutic indications for EUS can be broadly categorized as ductal access procedures (pancreatic or biliary), therapeutic drainage procedures (pancreatic fluid collection or gallbladder), endohepatology procedures (liver biopsy sampling, portal pressure assessment, or coil placement for varices), luminal anastomosis procedures (transgastric access, gastroenterostomy), and other (celiac plexus block [CPB], fiducial placement, or ablation) (13–18). The novel nature of these interventional EUS procedures and the associated nascent literature, which likely provides imprecise estimates of AEs, underscores the importance of thoughtful patient selection, multidisciplinary collaboration, and requisite endoscopic expertise. Performing these procedures only when indicated and when the risk-to-benefit ratio is favorable is particularly important.

2. Frequency with which informed consent is obtained, including a specific discussion of the risks associated with EUS, and the consent is documented
Level of evidence: 3
Performance target: >98%
Type of measure: Process

The document on quality indicators common to all endoscopic procedures (5) outlines the fundamental elements of appropriate informed consent, including those that pertain to sedation-related AEs. In addition to these fundamental elements, the AEs unique to EUS such as pancreatitis, infection, bile leak, or perforation, depending on the nature of the EUS, should be discussed and documented. Routine diagnostic EUS confers well-recognized and widely published risks. In contrast, the risks associated with the more novel interventional EUS procedures are becoming increasingly recognized and published in the literature.

Discussion. A 2022 ASGE document focusing on AEs of EUS divides procedures into 5 categories, each with their unique AEs: routine (diagnostic) EUS with or without FNA or fine-needle biopsy sampling (FNB), EUS for pancreatic fluid collection management, EUS-guided biliary (EUS-BD) and gallbladder drainage (EUS-GBD), EUS-guided CPB (EUS-CPB) or celiac plexus neurolysis (EUS-CPN), and other EUS-guided techniques including variceal management and EUS-guided gastroenterostomy (EUS-GE). The AE data presented in this document and others should form the basis of the risk-to-benefit discussion and consent process before EUS (19). When appropriate, the possibility of pivoting to an EUS-guided salvage approach after unsuccessful or impossible ERCP, as well as the associated risks, should be discussed and documented in the informed consent.

3. Frequency with which prophylactic antibiotics are administered for appropriate indications
Level of evidence: 1B
Performance target: >98%
Type of measure: Process

Prophylactic antibiotics should be administered only in settings where existing evidence supports that they are beneficial.

Discussion. Since the last version of this EUS quality indicator document, emerging evidence has challenged the conventional approach to antimicrobial prophylaxis in patients undergoing EUS-FNA of pancreas cysts. A multicenter randomized controlled trial (RCT) from Spain enrolling 226 patients undergoing EUS-FNA of pancreatic cystic lesions found that the infection rates in the group receiving antibiotics vs placebo were similar (0.4% and 0.9%, $P > 0.05$) (20). A subsequently published systematic review and meta-analysis found no statistically significant benefit associated with prophylactic antibiotics for FNA-related infections (odds ratio 0.54; 95% CI 0.16–1.92; $P = 0.32$) (21). Another meta-analysis that included the aforementioned RCT and 5 retrospective studies comprising 1,706 patients also noted no difference in infection rates between patients who received antibiotics and those who did not (22). These data and others suggest that routine use of prophylactic antibiotics in this context is not justified (23,24). Prophylactic antibiotics should nevertheless be considered when aspirating cysts in immunosuppressed patients, for those with numerous comorbidities in whom infectious AEs might prove particularly dangerous, when cysts are incompletely aspirated, and when ascites is present (25,26).

Mediastinal cysts pose a unique challenge because endosonographers are typically taught to sample most lesions to obtain an accurate diagnosis. However, the cyst infection and mediastinitis risk should preclude routine sampling of these lesions because even antibiotic prophylaxis has not been shown to prevent such AEs (27). Similar caution must be applied to rectal US-guided FNA and FNB of lower tract lesions. Despite the high diagnostic yield, especially for subepithelial and perirectal lesions, the risk of infection, including perirectal abscess, is approximately 15% despite prophylactic antibiotics (28,29).

No studies have rigorously evaluated the use of prophylactic antibiotics for EUS-FNA of presumed inflammatory pancreatic fluid collections (ie, pseudocyst and walled-off necrosis) either for diagnosis or drainage. Additionally, data are limited on the use of prophylactic antibiotics before performing interventional EUS. However, a 2022 European Society of Gastrointestinal Endoscopy technical review on therapeutic EUS recommended prophylactic administration of an intravenous broad-spectrum antibiotic in all patients undergoing therapeutic EUS procedures (weak recommendation, low-quality evidence) (30). The rationale for antibiotics in this context was based on historical data from interventional radiologic and surgical procedures wherein the peritoneum is breached during transumural therapeutic intervention (31,32). Table 4 summarizes the recommendations for prophylactic antibiotic use in EUS.

4. Frequency with which EUS is performed or supervised by an endosonographer who is fully trained and appropriately credentialed to perform that particular procedure
Level of evidence: 3
Performance target: >98%
Type of measure: Process

To ensure a high-quality procedure, EUS should only be performed by an individual with the requisite technical and cognitive skills, according to objective measures of competency. Competency is typically achieved through subspecialty training programs but can also be the result of postgraduate training pathways.

Table 3. Established and evolving indications for EUS

Diagnostic indications
Staging of GI tract malignancies
Evaluating abnormalities of the GI tract wall or adjacent structures
Evaluation of abdominal or pelvic abnormalities
Tissue sampling of lesions within or adjacent to the wall of the GI tract
Evaluation of abnormalities of the pancreas, including masses, pseudocysts, and chronic pancreatitis
Evaluation of patients at increased risk of pancreatic cancer
Evaluation of abnormalities of the biliary tree
Evaluation for perianal and perirectal disorders (anal sphincter injuries, fistulae, abscesses)
Therapeutic (interventional) indications
Treatment of symptomatic pseudocyst and walled-off necrosis by creating a gastroenteric–cyst communication
Providing access into the extrahepatic bile ducts or pancreatic duct, either independently or as an adjunct to ERCP
Providing access into the intrahepatic bile ducts independently with ability for decompression (EUS-guided drainage)
Treatment of acute cholecystitis or symptomatic cholelithiasis (gallbladder drainage)
EUS-directed transgastric access for altered anatomy
EUS-guided gastrojejunostomy
Tissue sampling of the liver
EUS-guided portal pressure assessment
EUS-guided coil placement for gastric varices
EUS-guided ablation
Celiac plexus block or neurolysis
Placement of radiologic (fiducial) markers into tumors within or adjacent to the wall of the GI tract

GI, gastrointestinal; ERCP, endoscopic retrograde cholangiopancreatography; EUS, endoscopic ultrasound.

Discussion. The most traditional route for acquiring focused technical and cognitive skills in EUS is the pursuit of an additional, advanced endoscopy (fourth-year) fellowship. Advanced endoscopy applicants and training positions have steadily increased over the years (33). Akin to other newer techniques, such as peroral endoscopic myotomy, endoscopic submucosal dissection, and endoscopic bariatrics, the question remains whether EUS can be learned and performed by individuals at a point in their career that precludes an additional year of advanced endoscopy fellowship. Expert opinion suggests that short weekend courses and training in animal models do not provide a permit to start independently performing these newer endoscopic procedures on humans (34).

Structured EUS training programs have been developed throughout the world. A successful program in Germany includes four 3-day training courses incorporating didactics, video sessions, live demonstrations, and hands-on experience (35). The ASGE has developed a diagnostic EUS course designed for the practicing physician and incorporates virtual learning modules, didactics, a hands-on animal laboratory, and a proctorship. The ASGE and other international consensus groups have developed a core curriculum for performing basic EUS (36–38). Tools to assess these basic skills outside of an apprenticeship model are in development (39).

Although training pathways are becoming more widespread, proper assessment of competency in basic and advanced EUS skills can be harder to measure and ensure. Adequate training now focuses on competency-based thresholds in lieu of an absolute number of EUS procedures needed to become "EUS-trained." However, for a rough estimate, Wani et al (40) studied advanced endoscopy training programs to define the number of procedures required by an "average" advanced endoscopy trainee to achieve competence in technical and cognitive EUS applying a cumulative sum analysis to create a learning curve for each trainee. The average trainee achieved competency at 225 cases.

Regardless of the fellowship or postgraduate training pathway, objective measures of competency should be achieved and documented before credentialing for EUS. Indeed, ASGE credentialing guidelines recommend that objective criteria and direct observation, not simply the number of procedures performed in training or practice, should be used to define competency (41). To date, there are no formal credentialing standards even for ERCP, although applying such standards may be forthcoming by the ASGE in the near future.

INTRAPROCEDURE QUALITY INDICATORS

5. Frequency with which the visualization of relevant structures, specific to the indication for the EUS, is documented

Level of evidence: 3

Performance target: >98%

Type of measure: Process

To maximize the quality of care associated with EUS, basic interpretation and description of wall layer of origin, lesion extension to surrounding structures, size, sonographic features, and pertinent positives and negatives relevant to the indication should be documented. Additionally, representative sonographic images should be obtained.

Discussion. Particular attention should be paid to the following descriptions and documentation:

1. In the setting of mediastinal lesions, documentation of the anatomic level of the lesion (A-P window, aortic arch, subcarinal) and the relationship to adjacent structures (azygous, aorta).
2. In the setting of esophageal cancer, description of location within the esophagus, measurements from the incisors of the proximal and distal extent of the tumor, whether there is obstruction, documentation of T staging, locoregional adenopathy, and the presence or absence of celiac adenopathy and left liver lobe involvement when feasible. In obstructing esophageal cancers, there is little clinical value and added risk associated with dilation solely for the purpose of satisfying this quality indicator (42).
3. For SELs, evaluation of the echogenicity, wall layer of origin, and other layers involved (see quality indicator 7, below).
4. In evaluating for pancreaticobiliary disease, visualization of the entire pancreas along with the pancreatic duct and its size. Key landmarks including the vasculature should be identified and documented in relation to the lesion. The presence or absence of worrisome features should be documented when describing pancreatic cysts. Changes of chronic pancreatitis with reference to the Rosemont criteria should also be documented. Recognizing aberrant or abnormal biliary anatomy including choledochal cysts, biliary dilation, choledocholithiasis and other pathology is important.
5. For lower GI tract indications such as rectal cancer staging; location of the tumor and visualization of the surrounding structures such as the iliac vessels, genitourinary organs, and sphincter muscles; and surrounding lymphadenopathy should be documented.

A high-quality EUS examination often parallels successful completion and documentation of the metrics (with accompanying images) identified in the ASGE training committee EUS core curriculum (36). These elements and those identified above form the crux of a high-quality examination and serve as the scaffolding for quality indicator in EUS. Photographic documentation of normal and abnormal findings should mirror the text in the findings.

6. Frequency with which luminal GI cancers are staged using the American Joint Committee on Cancer (AJCC) and Union for International Cancer Control TNM staging system

Level of evidence: 3

Performance target: >98%

Type of measure: Process

The adoption of EUS as a useful clinical tool was initially predicated on its superior operating characteristics for luminal cancer staging relative to other diagnostic modalities that do not provide as detailed imaging of the GI tract wall. Valuable information that is critical to the diagnosis, management, and prognosis of luminal GI cancers can be gleaned by EUS, generally performed with a radial echoendoscope in this context.

Discussion. Luminal cancers should all be notated according to the TNM classification system (43). EUS is highly accurate for the primary staging of esophageal cancer and can delineate T1 lesions from T2-4 lesions (44). Overall, EUS T-stage accuracy in esophageal cancer (71%–92%) has been shown to be superior to computed tomography (CT) (42%–60%) (45–47). EUS, however, is not as accurate for assessing nodal disease. Unfortunately, other imaging modalities including CT and positron emission tomography are also limited for N-staging. A study of 112 patients that evaluated the operating characteristics of contrast-enhanced CT, EUS, and positron emission tomography for N-staging of esophageal cancer found that CT/positron emission tomography was more likely to under-stage nodal disease than EUS (48). Despite the low accuracy for N-staging in this study (77%), the operating characteristics of EUS remain on par with other imaging modalities.

EUS is also highly accurate in distinguishing early from advanced gastric cancer based on depth of invasion (49). One study, however, found that a dedicated stomach protocol CT and EUS had similar accuracy for T staging in gastric cancer (near 70%) but EUS performed better in differentiating T1 vs T2 disease (50). Furthermore, this study showed that when a stomach protocol CT and EUS were used in conjunction, there was a nearly 85% accuracy for T3-4 disease (50).

Rectal US is also accurate for T (80%–90%) and N-staging (70%–80%) of rectal cancers (51). In fact, for early-stage rectal cancer, available data indicate that rectal EUS is very useful in determining patients suitable for a minimally invasive vs a more radical operation (52). However, for the most part, MRI has supplanted rectal US for the primary staging of rectal cancer given its superior ability to assess the mesorectal fascia, which has important implications in planning for a total mesorectal excision (53). Regardless, when EUS is performed for rectal and other luminal cancers, staging should be performed according to the AJCC and Union for International Cancer Control TNM staging system. When applying TNM staging using EUS for lesions suspected to be superficial, it is important to consider that pathologic evaluation of resection specimens is superior to EUS for depth of invasion (ie, distinguishing T1a vs T1b or T2 cancers) (54–56).

7. The frequency with which the echogenicity and wall layer of origin of subepithelial masses are documented (priority indicator)

Level of evidence: 3

Performance target: >98%

Type of measure: Process

EUS is an essential tool in the evaluation of SELs. EUS can establish whether a SEL is extrinsic or intrinsic and, if intrinsic, identify the echogenicity and layer(s) of origin. EUS allows high-frequency imaging that can help delineate the gastric wall layers and aid in the diagnosis of gastric SELs (57). Up to 30% of suspected SELs are extrinsic (58,59), and endoscopy alone is often insufficient for differentiation. EUS is far superior to endoscopy for determining the origin of a SEL (extrinsic vs intrinsic as well as layer of origin) (60). In an international multicenter study, the sensitivity and specificity of correctly differentiating an intramural from an extramural lesion with endoscopy alone were 87% and 29%, respectively. EUS improved the sensitivity and specificity to 92% and 100%, respectively (58).

Once a lesion is determined to be intrinsic, noting the echogenicity and wall layer of origin is imperative because this is critical in narrowing the differential diagnosis. For example, a hypoechoic lesion that originates from the muscularis propria is likely to be a leiomyoma or GI stromal tumor. Conversely, a hyperechoic lesion arising from the submucosa is most likely a lipoma. In a case series that included 53 gastric SELs evaluated by both EUS and CT before laparoscopic wedge resection, the overall diagnostic accuracy of EUS (without FNA) was 64% compared with only 51% with CT (61). Additionally, accurate determination of the wall layer of origin informs whether resection is feasible and oncologically beneficial, and if so, whether endoscopic or surgical removal is more appropriate. For example, endoscopic resection may be feasible if the lesion is limited to the submucosa, but full-thickness endoscopic resection or surgery is necessary if the lesion originates from the muscularis propria.

High-frequency probes allow for very detailed imaging of the gastric wall layers and may further improve the diagnostic accuracy of EUS for SELs (62). One retrospective study suggested a higher diagnostic accuracy (92.6%) when using high-frequency probes (63). However, a more recent retrospective study evaluating a 20-mHz probe reported only an 80.1% diagnostic accuracy compared with pathology for upper GI SELs, with the highest accuracies for lipomas and pancreatic rests, but only 63% accuracy for GI stromal tumors (64).

Although EUS is superior to endoscopy for morphologic assessment, tissue acquisition further increases the diagnostic accuracy in patients with solid nonlipomatous SELs. Indeed, using pathologic criterion standards, EUS imaging without tissue acquisition has only an approximately 43%–50% diagnostic accuracy, 64%–80% sensitivity, and 77%–80% specificity for differentiating benign from malignant SELs (58,60,65–67). Fine-needle sampling increases this accuracy to 75%–100% (68,69).

When performing EUS-guided sampling for SELs, EUS-guided FNB (EUS-FNB) or EUS-FNA with rapid onsite evaluation (ROSE) is recommended (70). In many SELs, immunohistochemical and biomarker staining and assessment of mitotic figures are crucial. One meta-analysis of 669 patients from 10 studies (including 6 randomized trials) compared EUS-FNA with EUS-FNB of SELs (71). FNB yielded higher rates of adequate samples and histologic cores, yielded greater diagnostic accuracy, and required fewer needle passes than FNA. However, the 2 techniques were similar in the above outcomes when ROSE was available. Interestingly, another meta-analysis examining solely upper GI SELs found no difference in diagnostic yield comparing FNA, Tru-cut (Cook Medical, Winston-Salem, NC) core biopsy sampling, and FNB regardless of needle size or use of ROSE (72).

8. The frequency with which a diagnostically adequate EUS-guided liver biopsy sample is obtained
Level of evidence: 1B
Performance target: >85%
Measurement type: Outcome

EUS-guided liver biopsy sampling (EUS-LB) has emerged as an alternative to percutaneous, transvenous, or laparoscopic biopsy sampling. Advantages of EUS-LB include the ability to visualize the position of the needle in real time with Doppler capability to aid in avoiding vasculature, sampling of both lobes of the liver, and allowing the performance of concomitant endoluminal evaluation, pancreaticobiliary intervention, and EUS-guided portal pressure measurement.

Discussion. The American Association for the Study of Liver Disease and the Royal College of Pathologists define an adequate sample as the combination of a total specimen length >2 cm and >11 complete portal triads (73–75). Because this is often difficult to achieve, many studies have redefined adequacy as a total specimen length of 1.5 cm and at least 6 complete portal triads (76,77).

Existing literature suggests that the specimen adequacy of EUS-LB is in excess of 90% when defined as complete portal triad >6 and total specimen length >15 mm. For example, a meta-analysis demonstrated that when using this standard, EUS-LB achieved histologic success (diagnostic yield) in 93.9% of cases (76). Another meta-analysis using the same standard, which included 33 studies, noted that pooled rates of diagnostic yield, specimen adequacy, and AEs were 95%, 84%, and 3%, respectively (77). A third meta-analysis concurred that EUS-LB is efficacious and safe (78). However, these data may be biased because most studies published thus far have been performed by experienced endoscopists who perform these procedures in high-volume centers. To account for this potential bias in the adequacy of EUS-LB specimens, a performance target of >85% has been assigned to this quality indicator, although we anticipate that this will increase as EUS-LB diffuses more broadly.

When compared with other approaches, EUS-LB appears to be equivalent or superior to the transjugular approach (79–83). However, the data comparing EUS-LB with percutaneous biopsy sampling are conflicting with no clear winner, although EUS-LB makes sense when accompanying endoscopic screening or surveillance, ERCP, or EUS-guided portal pressure gradient measurement.

From a technical perspective, 19-gauge fine-needle biopsy sampling (typically using a forked or Franseen geometry) is superior to traditional FNA in terms of specimen length (2.09 vs 1.47 cm) and number of complete portal triads (42.6 vs 18.1) (77), and thus FNB needles are favored for EUS-LB. These same investigators, however, found that AE rates were higher after FNB compared with FNA sampling (6% vs 1%, $P = 0.028$). Among FNB needles, a prospective single-center study demonstrated that the 19-gauge Franseen needle had statistically higher diagnostic adequacy compared with the fork-tip FNB needle (84). One of the aforementioned meta-analyses also showed that Franseen biopsy needles had a higher diagnostic yield (99% vs 88%, $P = 0.047$). Most studies show that flushing the needle with heparin and using a wet-suction technique compared with dry suction increases the yield of EUS-LB (85–88).

9. The frequency with which a pancreatic mass ≥10 mm is identified with EUS
Level of evidence: 2C
Performance target: ≥90%
Type of measure: Outcome

Cross-sectional imaging is typically the initial diagnostic test to detect most pancreatic masses. However, ample data highlight the shortcomings of high-resolution CT in its ability to detect pancreatic tumors <2.0 cm. The utility of EUS is magnified for these difficult to detect and characterize lesions.

Discussion. EUS is highly accurate for detecting pancreatic cancer, with a sensitivity of 93%–100% and specificity of 53%–100% (89,90), and is particularly helpful for identifying small pancreatic cancers, for which it is superior to CT (90). This is highlighted in a systematic review and meta-analysis of 206 subjects with a clinical suspicion of pancreatic cancer but indeterminate findings on multidetector CT (91). In this study, EUS had a pooled sensitivity of 85% (95% CI 69–94) and specificity of 58% (95% CI 40–74) for diagnosing pancreatic malignancy, where the mean size of the lesion diagnosed by EUS was 2.1 ± 0.12 cm (91). These and similar data, as well as the expert opinion of the task force, justify a performance target of >90% for endosonographic identification of pancreatic masses ≥ 10 mm.

Contrast-enhanced EUS is widely used in Europe and the Far East and allows visualization of blood flow within a tumor, which can help identify pancreatic cancer (92). A meta-analysis of 16 studies with 1,325 subjects found that contrast-enhanced EUS had a pooled sensitivity of 93% (95% CI 91–94) and specificity of 84% (95% CI 80–87) for detecting pancreatic adenocarcinoma (93). The widespread availability of technology may contribute to a higher performance target for this indicator in the future. EUS-guided elastography, which assesses tissue stiffness in real time, may increase the sensitivity of EUS for detecting malignancy (94,95). This technology is playing an emerging role in diagnosing malignancy in the setting of chronic pancreatitis, in which the sensitivity of EUS-FNA is lower (96,97).

10. The frequency with which pancreatic mass measurements, vascular and lymph node involvement, liver metastases, and ascites are evaluated and documented

Level of evidence: 3

Performance target: ≥90%

Measure type: Process

Table 4. Specific indications for prophylactic antibiotics in EUS

Indication	Need for prophylactic antibiotics
Pancreatic solid lesion	Not indicated
Pancreatic cystic lesion	Not routinely indicated
Pancreatic pseudocyst	Case by case
Pancreatic necrosis (FNA or debridement)	Should administer
Interventional EUS	
EUS-guided pancreatic fluid collection	Strongly consider
EUS-guided gallbladder drainage	Strongly consider
EUS-guided biliary drainage	Strongly consider
EUS-guided gastroenterostomy	Strongly consider
EUS-directed transgastric ERCP	Strongly consider

Few of the recommendations are based on consensus expert opinion.
ERCP, endoscopic retrograde cholangiopancreatography; EUS, endoscopic ultrasound; FNA, fine needle aspiration.

A proper examination when interrogating a pancreatic mass should include accurate measurement of the lesion, thorough inspection of extrapancreatic tissues to identify the presence of lymph nodes according to their station as designated by the AJCC (peripancreatic, celiac, porta hepatis, gastrohepatic ligament, aortocaval), assessment of the presence and extent of vascular invasion (length as well as luminal compromise, presence of tumor thrombus), and assessment for liver metastasis and ascites. Each nodal region corresponds to a designated station based on the AJCC nomenclature; for example, the celiac node would correspond to station 9 and the peripancreatic lymph node stations would correspond to stations 17 and 18 (98).

Discussion. In current clinical practice, multidetector CT and MRI are the primary tools for identifying and staging pancreatic cancer. EUS, however, plays a complementary and important role, particularly in identifying pathology that might upstage the cancer and thus change patient management. Meta-analyses show that EUS has a pooled sensitivity of between 66% and 85% with a specificity of between 89% and 94% for detecting vascular invasion (99,100). Small older studies have reported that EUS is particularly sensitive for identifying portal vein involvement with greater than 80% sensitivity. Involvement of celiac, peripancreatic, porta hepatis, gastrohepatic ligament, and aortocaval lymph nodes should be evaluated, with a metaanalysis showing that EUS had a sensitivity of 69% and specificity of 81% for detecting lymph node involvement (99).

EUS cannot reliably identify all distant metastases but is useful for finding small liver metastases and malignant ascites not identified on CT (101,102). A retrospective study of 426 subjects with pancreatic cancer found that EUS detected very small liver metastases with a diagnostic accuracy of 93.4%, which increased to 98.4% when contrast-enhanced EUS was used (102). A recent prospective study of 730 individuals, two-thirds of whom had pancreatic cancer, reported that EUS identified liver metastases in 16% of subjects, 22% of which were not seen on CT (103). Based on this literature demonstrating the value of EUS in this context, endosonographers should evaluate and document the maximum pancreatic mass diameter, vascular and lymph node involvement, and liver metastases and ascites in 90% or more cases.

11. The frequency with which a diagnostic specimen is obtained by EUS-guided sampling of a malignant pancreatic mass (priority indicator)

Level of evidence: 1C+

Performance target: ≥87%

Type of measure: Outcome

Pancreatic mass lesions can occur because of several different benign and malignant etiologies including mass-forming chronic pancreatitis, autoimmune pancreatitis, pancreatic adenocarcinoma, pancreatic neuroendocrine tumors, metastases to the pancreas, and lymphoma. A major objective of EUS in evaluating pancreatic masses is to confirm the etiology by obtaining a pathologic diagnosis through guided FNA or FNB.

Discussion. A meta-analysis evaluating the diagnostic accuracy of EUS-FNA in 33 studies (4,984 subjects) found a pooled sensitivity of 85% (95% CI 84–86) and specificity of 98% (95% CI 0.97–0.99) for malignant cytology (104). When only 22 prospective studies were included, the pooled sensitivity increased to 87% (95% CI 86–88), with no change in specificity (104). Another meta-analysis evaluating 41 studies (4,766 subjects) reported similar findings, with a pooled sensitivity of 86.8% (95% CI 85.5–87.9) and specificity of 95.8% (95% CI 94.6–96.7) (105). Based on these data, the performance target for the diagnostic yield of EUS-guided sampling of pancreatic masses has been set at 87%. Several technical approaches may maximize tissue acquisition and diagnostic yield. The first pertains to needle beveling and whether FNB is superior to traditional FNA. Twenty-nine RCTs and 5 meta-analyses compared the performance characteristics of FNA vs FNB for solid pancreatic masses (106–109). Four meta-analyses (106,108–110), which included up to 26 RCTs, reported a higher diagnostic accuracy of FNB compared with FNA. Van Riet et al (108) reported a pooled diagnostic accuracy of 87% (95% CI 85–89) for FNB compared with 80% (95% CI 78–82) for FNA. Although these 4 meta-analyses found FNB was superior, in a subanalysis that included only RCTs in which ROSE was available, FNB was not significantly superior to FNA (109). In contrast to these 4 meta-analyses, Facciorusso et al (107) performed a meta-analysis of 27 RCTs (2,711 subjects) and found no difference between EUS-FNA and EUS-FNB, with a pooled sensitivity of between 89.9% and 90.8% for 25-gauge and 87.9% and 94.7% for 22-gauge FNA and FNB needles, respectively. The reasons for the differences between these meta-analyses include different inclusion criteria, inclusion of newer studies, development of new FNB needles, and significant heterogeneity in the studies with the quality of the evidence graded as low. However, overall, it appears that FNA with ROSE is an acceptable approach, but FNB should be favored where ROSE is not available or when core tissue is required.

Another consideration is the effect of needle size on diagnostic accuracy. Five meta-analyses compared the performance of different needle sizes (107,111–114). Three meta-analyses, which include up to 26 RCTs and 2,711 subjects, found no difference in diagnostic accuracy between 22gauge and 25-gauge needles (107,111,112). In contrast, 2 meta-analyses, which included 8 (113) and 11 (114) studies, respectively, and evaluated up to 1,292 subjects found a higher diagnostic accuracy associated with 25-gauge needles, with a pooled sensitivity of 92%–93% compared with a sensitivity of 85%–88% for the 22-gauge needle.

The 87% performance target is a higher threshold than previous targets. Based on refinements in tissue acquisition, both in FNA and core biopsy needle designs, as well as in techniques (slow pull, fanning, suction, vs no suction), this new target is supported by evolving evidence.

12. Frequency with which technical success in EUS-guided pancreatic fluid collection drainage (PFCD) is achieved (priority indicator)
Level of evidence: 1B
Performance target: ≥92%
Type of measure: Outcome

EUS-PFCD has become widely accepted as an effective and often primary drainage method in patients who develop fluid collections, either from a pseudocyst or walled-off necrosis, because of pancreatitis or after surgical resection of the pancreas (115–117). Earlier studies reported on a multistep, multidevice approach to place plastic stents or fully covered metal stents into the cavity; however, the more recent diffusion of an electrocautery-enhanced lumen-apposing metal stent (LAMS) system has increased the rate of technical success, which now ranges between 92% and 100% for collections that are amenable to transmural drainage (118–120).

Discussion. Growing evidence confirms that EUS-PFCD is associated with high technical and clinical success rates (121,122). In a large prospective, cohort study, the technical success rate of EUS-PFCD with LAMSs was 91% with pancreatic fluid collection resolution in 93% (122).

Several RCTs and meta-analyses have compared outcomes between EUS-PFCD and surgical or percutaneous drainage (117,120,123–127). In aggregate, these studies demonstrate similar success rates between modalities but less AEs, especially pancreaticocutaneous fistulae, associated with EUS-PFCD. This comparative literature affirms that EUS-PFCD has technical success rates of 92% and clinical success approaching 98%, justifying the performance target of ≥92%.

13. Frequency with which technical success in EUS-GBD is achieved
Level of evidence: 2B
Performance target: >90%
Type of measure: Outcome

The indications EUS-GBD are primary treatment of acute cholecystitis or symptomatic cholelithiasis in patients at high surgical risk (128), gallbladder drainage after unsuccessful or impossible percutaneous or other drainage method, conversion of percutaneous cholecystostomy drains to internal drainage in patients who are not candidates for cholecystectomy, and treatment of distal biliary malignant obstruction when ERCP or direct bile duct drainage is not feasible (129–131). Emphysematous cholecystitis, contained gallbladder perforations associated with cholecystitis, and large amounts of ascites warrant extra caution. A multidisciplinary discussion before EUS-GBD is advisable in patients who are not surgical candidates at the time of presentation but have the potential for future cholecystectomy, because the resultant cholecystogastrostomy or duodenostomy may complicate the subsequent operation. This dogma has been challenged by several recent studies that reported favorable outcomes in patients undergoing elective cholecystectomy after EUS-GBD (128,132). In fact, these same investigators conducted a multicenter, international study of patients undergoing cholecystictomy for acute cholecystitis after either a percutaneous transhepatic gallbladder drainage (PT-GBD) or EUS-GBD and found no difference in surgical technical success (132).

Discussion. The original description of EUS-GBD, requiring multiple steps to deploy plastic or tubular self-expandable metal stents, was prone to AEs because of the potential for wire loss and stent maldeployment and the suboptimal nature of these prostheses. Electrocautery-enhanced LAMSs have overcome these limitations, making the procedure safer and gaining the U.S. Food and Drug Administration approval for this indication. Other forms of stents have thus fallen out of favor for EUS-GBD.

Studies and meta-analyses comparing clinical outcomes between EUS-GBD, ERCP-GBD (with placement of a transpapillary gallbladder stent), and PT-GBD (133–137) showed that EUS-GBD and PT-GBD have comparable technical success rates of 95.3% (95% CI 92.8–96.9) and 98.7% (95% CI 98.0–99.1), respectively. Both techniques had better technical success than ERCP-GBD (83%; 95% CI 80.1–85.5). EUS-GBD had the highest clinical success of 96.7% (95% CI 94.0–98.2) (133). Overall, it does appear that with improved devices and experience, the technical and clinical success rates are in the range of 95% but approach 98%–100% in some studies (135,137).

However, because existing publications largely originate from expert centers and the early literature for any novel procedure tends to exaggerate success, the performance target for EUS-GBD has been set at a more conservative 90%, which still compares favorably with the expected success of ERCP-GBD, for example. However, the long-term clinical implications and benefit of EUS-GBD vs ERCP-GBD remain to be formally compared and determined.

14. Frequency with which technical success in EUS-BD is achieved
Level of evidence: 1B
Performance target: >85%
Type of measure: Outcome

The indications for EUS-BD are failed ERCP in patients with an accessible ampulla, inaccessible ampulla in patients with gastric outlet obstruction (GOO), and difficult to access ampulla in patients with surgically altered anatomy. Contraindications include tumor infiltrating the intended puncture site along the luminal surface, inability to visualize the bile duct, massive ascites, coagulopathy, and standard contraindications to endoscopy and sedation (138).

Currently, EUS-BD is guideline recommended over percutaneous transhepatic biliary drainage for rescue treatment of extrahepatic malignant obstruction after unsuccessful or impossible ERCP if expertise is available (139). Emerging evidence also supports the potential use of EUS-BD as a primary drainage for distal malignant obstruction with outcomes comparable with ERCP (140,141). For benign obstruction, however, a multidisciplinary approach remains particularly important, and case selection should be individualized after thoughtful discussion.

Drainage of the bile duct can be achieved using the rendezvous technique, wherein EUS is used to advance a wire through the papilla or biliary–enteric anastomosis to facilitate subsequent ERCP, direct transmural stent placement (including hepaticogastrostomy [HGS] and choledochoduodenostomy), or antegrade stent placement across the stricture. These interventions are almost always applied in the context of malignant biliary obstruction; however, EUS-guided rendezvous technique is often used for access after unsuccessful ERCP regardless of indication (142,143).

Discussion. The overall technical and clinical success of EUS-BD in the literature ranges from 90% to 96% with AEs ranging from 8% to 27%, comparable with those observed with ERCP for similar indications (138,144–146). However, 1 RCT showed that ERCP had more frequent AEs, especially postprocedure pancreatitis (147). A multicenter study comparing endoscopic placement of expandable metal stents for malignant distal common bile duct obstruction by ERCP or an EUS-guided approach found that postprocedure pancreatitis rates were higher in the ERCP group (4.8% vs 0%, $P = 0.59$) (148). Several studies have been conducted to compare EUS-guided (EUS-HGS) with EUS-guided choledochoduodenostomy. These data suggest that both techniques have comparable technical and clinical success rates. However, Ogura et al (149) reported higher clinical success and fewer AEs with EUS-HGS. Again, because published studies are typically performed at expert centers, the performance target for this quality indicator in general practice was set at 85%.

Several techniques proposed by experts are believed to increase technical success and minimize AEs. These techniques include using EUS guidance throughout the entire procedure until stent deployment instead of alternating between EUS and endoscopic visualization, using the liver impaction technique (pulling the needle back into the liver parenchyma, allowing the needle tip to become impacted by the parenchyma, and the angle between the needle and guidewire becomes obtuse, avoiding wire kinking or shearing while directing the wire in the desired direction), using either a noncautery dilator or small size (6F) cystotome to avoid bleeding, using fully covered metal stents to avoid bile leak, and intrascope channel stent deployment to avoid stent maldeployment (150–152). For EUS-HGS, either fully or partially covered metal stents are preferred, whereas either tubular metal stents or LAMSs are appropriate for EUS-guided choledochoduodenostomy. When LAMSs are used for this indication, sufficient working space in the bile duct is necessary to allow complete deployment of the intraductal flange. The requisite distance varies according to the size of the prosthesis but should be accounted for in every case to minimize maldeployment (30,141).

15. Frequency with which technical success in EUS-GE is achieved
Level of evidence: 2C
Performance target: >85%
Type of measure: Outcome

EUS-GE has recently been accepted as an alternative method for the treatment of malignant GOO. Additionally, the indications are expanding to include treatment of benign obstructions, including afferent loop syndrome, with favorable outcomes reported by experts. As with EUS-GBD, this procedure initially required multiple steps to puncture the small bowel, expand the tract, and place a stent into a suitable loop of jejunum. With the advent of the electrocautery-enhanced LAMSs, the procedure is now much more efficient and can be achieved by 3 main approaches: direct, through a nasobiliary catheter, or using an assist device (153).

Discussion. EUS-GE is most widely accepted as an alternative to enteral stent placement or surgery for malignant, unifocal GOO (154,155). However, EUS-GE can also be used for conditions that cause benign obstruction of the foregut, such as chronic pancreatitis, peptic ulcer disease, or superior mesenteric artery syndrome (156). Life expectancy in patients with malignant GOO was cited as one of the main decision drivers of when to offer EUS-GE in a recent international practice survey (157). Several studies have demonstrated the clinical superiority of EUS-GE (using LAMSs) over enteral stent placement and equivalent technical success to surgical gastrojejunostomy. Additionally, emerging data suggest that the time to resumption of oral intake is shorter with EUS-GE compared with surgery (158–163). In both malignant and benign conditions, the technical success rate of EUS-GE is reported to be in the range of 92%–100% (158,159,161,164,165). With ongoing improvement in techniques and devices, the technical success rate consistently approaches 95% in centers with established expertise (166). As above, the lower performance target is to account for bias inherent to the early literature of any procedure.

The 2 most common techniques involve flooding of the small bowel with saline solution and contrast directly followed by direct puncture with the LAMS or inserting a nasobiliary catheter for continued irrigation to distend a suitable loop of bowel for EUS-guided stent placement. The third technique involves the use of dedicated assist devices, which are expected to further improve the technical success and safety of these procedures. EUS-guided balloon-occluded gastroenterostomy bypass, for example, achieved a 95% technical success rate and decreased the procedure times with an overall clinical success of 89.2% (161).

16. Frequency with which technical success in EUS-directed transgastric ERCP (EDGE) is achieved
Level of evidence: 1C
Performance target: >92%
Type of measure: Outcome

Patients who require access to the excluded stomach or duodenum after Roux-en-Y gastric bypass can undergo EUS-guided gastrogastrostomy or jejunogastrostomy using a LAMS. This procedure was originally conceived to facilitate ERCP (EDGE) but has evolved into a pathway for any endoscopic intervention in the excluded foregut (167). Given the risks, need for multiple procedures, and the possibility of persistent gastrogastric or gastricjejunal fistula, discussion with patients and treating clinicians about the merits and disadvantages of this approach relative to laparoscopicassisted ERCP and enteroscopy-assisted ERCP is essential.

Discussion. Comparative data with other techniques such as laparoscopic-assisted ERCP and enteroscopy-assisted ERCP have found similar rates for technical and clinical success but a lower rate of AEs and procedure time (168–171). A persistent fistula rate once the LAMS is removed is a concern, and various factors (including dwell time) have been attributed to this (172). Endoscopic closure of the fistula at the time of stent removal appears to reduce this risk, and other technical factors, such as the location of the fistula creation, are being studied for this purpose. The 3 largest studies have shown technical success rates ranging between 96% and 99% (168,171,173). This rate of success is affirmed by 2 recent systematic reviews showing a pooled technical success of 96%–99% (174,175).

POSTPROCEDURE QUALITY INDICATORS

17. Frequency with which a complete report is created
Level of evidence: 3
Performance target: >98%
Type of measure: Process

An accurate, detailed, and timely procedure report is required for every EUS procedure.

Discussion. Two publications, the QUOREUS study and the Forum for Canadian Endoscopic Ultrasound Working Group, explored the quality of reporting in EUS as well as the standard reporting elements for high-quality EUS (7,176). Fusaroli et al (176) conducted an online survey of 171 international EUS experts, showing strong agreement that standardization is essential for high-quality reporting. Along these lines, a group of experts at the Forum for Canadian Endoscopic Ultrasound meeting in 2019 compiled a list of essential reporting elements to ensure best practices in EUS. When considering these 2 documents, the ACG-ASGE document on quality indicators common to all endoscopic procedures (5), and additional expert opinion, elements outlined in Supplementary Appendix 1 (see Supplementary Digital Content 1, http://links.lww.com/AJG/D643) constitute a high-quality EUS report. Adopting and complying with a standardized reporting framework, as outlined above, allows for more efficient accurate measurement and benchmarking in EUS (177).

18. Frequency with which the incidence of AEs after diagnostic and interventional EUS are documented (priority indicator)
Level of evidence: 3
Performance target: >98%
Type of measure: Process

Capture and documentation of AEs is a critical aspect of measuring and improving quality in EUS, particularly to avoid under-recognized harm associated with the more novel interventional EUS procedures.

Discussion. Endosonographers have shown greater compliance with quality indicators with improvement in documentation of pre-, intra-, and postprocedure parameters (178). However, documentation of AEs remains suboptimal. One study found that 0% of procedure reports listed AEs on the EUS report (179). An international survey of 171 endosonographers demonstrated consensus on the need for standardization of EUS reports to include documentation of AEs (176).

To accurately document AE rates, delayed events that occur after the patient has been discharged from the endoscopy unit must also be captured. Reliance on notification of an AE by the patient or another provider is likely inadequate as 15%–45% of AEs go unrecognized or unreported (180,181). Instead, active efforts to contact patients and identify delayed AEs, ideally 2 weeks after the procedure, are recommended with the goal of accurate documentation of AEs. Compliance with this quality indicator will become more feasible with future integration of interoperable electronic health records, practice management systems, and endoscopy report writers, which will allow searchable data warehouses to identify delayed AEs. Additionally, patient-facing smartphone applications designed to allow seamless communication between the patient and endoscopy personnel may limit the administrative burden associated with these efforts.

19. Frequency with which AEs after diagnostic EUS, including EUS-FNA, EUS-FNB, EUS-guide fine-needle injection, and EUS-LB, occur
Level of evidence: 1C+
Performance target: Perforation <0.5%, infection <1%, acute pancreatitis <1%, clinically significant bleeding <1%, with clinically significant bleeding <5% after EUS-LB.
Type of measure: Outcome

AEs after diagnostic EUS, including fine-needle sampling and injection, are rare. Nonetheless, such AEs are well described, and efforts to track and minimize them are important in ensuring high-quality EUS practice.

Discussion. AEs can occur even after purely diagnostic EUS procedures. Factors unique to the echoendoscope (rigidity and optics) can lead to a slightly greater risk of perforation when compared with EGD. The distal 4-to 5-cm segment of the echoendoscope is more rigid than the tip of a standard endoscope, whereas the optics are oblique-viewing. Because of the optics, the insertion and advancement of these scopes, especially across bends, are semi-blind and predispose to slightly higher perforation rates than that associated with EGD (182). A large retrospective study of over 13,000 procedures found that esophageal perforations occurred in 0.02% of all EUS procedures (both cases with a radial echoendoscope), whereas gastric and duodenal perforations occurred at a rate of 0.06%. Perforations were more common with the linear echoendo-scope (183). In contrast, another prospective study of 4,800 patients reported an esophageal perforation rate of 0.06%, all occurring with the linear echoendoscope (184). Advanced trainee involvement, operator inexperience, advanced patient age, history of difficult esophageal intubation, esophageal malignancy, and cervical osteophytes have all been independently associated with higher perforation rates during EUS (184–187).

EUS-FNA can lead to AEs such as bleeding, infection, pancreatitis, and needle tract seeding (188). Bleeding can occur after either FNA or FNB but is usually self-limited and rarely requires further intervention. A large systematic review of over 10,900 patients undergoing EUS-FNA reported a pooled bleeding rate of 0.13% (189). Of note, neither the needle diameter nor the number of passes was associated with the incidence of bleeding after EUS-FNA (190). A study suggested there might be a slightly increased rate of bleeding with the use of an FNB needle, but the absolute risk still remains low (191).

Although incidental bacteremia can occur after EUS with and without FNA, its clinical significance has been questioned because none of the affected patients in study cohorts ever developed sepsis or had their clinical course altered as a result of the EUS-related bacteremia (192–194). Similarly, infections arising after EUS-FNA of pancreatic cysts are minimal; a meta-analysis of 40 studies that included over 5,000 patients undergoing EUS-FNA of pancreatic cysts reported a pooled infection rate of 0.4%, independent of antibiotic prophylaxis as discussed above (20,190). Although Levy et al (195) showed no increase in bacteremia after transrectal FNA, a more recent study (29) demonstrated substantial rates of infectious AEs after transrectal FNA of cystic lesions—but not of solid lesions—despite antibiotic prophylaxis. Caution and discussion with surgical colleagues is paramount before transrectal FNA of a cystic lesion.

Pancreatitis after EUS-FNA or EUS-FNB of solid and cystic pancreatic lesions is low, varying between 0.5% and 1.5%. Zhu et al (190) reported a pancreatitis rate of 0.92% across 40 studies in a meta-analysis of over 5,000 patients undergoing FNA or FNB of pancreatic cystic lesions. The use of microforceps biopsy sampling to sample the wall of the cyst appears to increase pancreatitis risk slightly, although this rate is not well defined given the limited number of related studies (196,197). More recent interventions such as EUS-guided fiducial placement and EUS-guided radiofrequency ablation may also increase the risk of pancreatitis, although the rate remains quite low overall (198–200).

Although needle tract seeding after fine-needle sampling of tumors is rare, the potential should be discussed with patients before performing FNA or FNB of malignant lesions. Facciorusso et al (201) conducted a meta-analysis of 13,000 patients across 10 studies and found that the pooled rate of needle tract seeding was 0.3%. Furthermore, they found no difference in metachronous peritoneal dissemination in patients undergoing EUS-guided tissue acquisition compared with cancer patients, who do not undergo such sampling. However, studies from Japan have suggested decreased survival after transgastric EUS-guided sampling of lesions when the needle tract is outside the surgical resection field (202,203). Another scenario of particular concern is EUS-FNA and EUS-FNB of a perihilar biliary stricture or mass because of the associated risk of transperitoneal needle tracking and consequent peritoneal seeding. Although existing data are conflicting (204,205), most transplantation protocols for cholangiocarcinoma consider prior transperitoneal sampling an absolute exclusion criterion, and thus EUS-FNA and EUS-FNB of the primary lesion should be discouraged. Sampling of lymph nodes, however, is diagnostically helpful and is not contraindicated because the presence of malignancy in such nodes precludes transplantation.

EUS-CPB and EUS-CPN have been used to mitigate chronic abdominal pain originating from chronic pancreatitis and pancreatic cancer, respectively. Transient diarrhea has been reported in 2% of patients after EUS-CPB and 10% of patients after EUS-CPN (206). This same review found that hypotension occurred in 2% of patients with CPB and 5% of patients after EUS-CPN. Although this overall risk profile is considered acceptable, particularly severe AEs after these techniques have been reported in case reports or series. Retroperitoneal abscess formation and empyema (207,208) as well as paraplegia and fatal AEs have been reported and deserve cautious consideration (209–211).

20. Frequency with which AEs related to interventional EUS procedures occur

Level of evidence: 2C

Performance target: Overall AEs of <10% for EUSPFCD, <20% for EUS-GBD, <25% for EUS-BD, <15% for EUS-GE and EDGE

Measure type: Outcome

Interventional EUS procedures encompass a spectrum of potential AEs, ranging from bleeding, double puncture, wire displacement, bile leak, stent misdeployment, perforation, pneumoperitoneum, and infection. Additionally, delayed AEs such as stent dislodgment, vascular injury leading to pseudoaneurysm and bleeding, and stent malfunction may arise. Procedures involving multistep techniques inherently carry a higher risk of AEs. Furthermore, certain AEs, including bile leak, wire displacement, and infection, exhibit higher incidence during biliary drainage procedures than in fluid collection drainage and luminal anastomosis (138,212–214). Consequently, performance targets vary across these procedures.

The overall AEs after PFCD with the use of LAMs range from 13% to 15%, with bleeding and stent misdeployment or dysfunction being the leading events (215). EUS-GBD has an overall morbidity range from less than 10% up to 20%, with a mortality of 3.9%, which is mainly related to ongoing sepsis (129,130). EUS-BD requires multiple steps and thus is most likely to have a higher rate of AEs. The overall AEs reportedly range from 3% to 39%

with bile leak as the most common (138,145,148,212,216–219). Mortality has been reported in 0.5% of cases (220–222). Stent misdeployment associated with perforation is the most common AE for EUS-GE and EDGE, accounting for 10% (213,214,223).

Discussion. Bleeding Bleeding risk varies between different procedures, and the risk can be minimized by using Doppler guidance before puncturing. The bleeding incidence is reportedly 4% after EUS-BD (138,212) and 8% after EUS-GBD (224). Technical refinements and a single-step device may minimize bleeding related to biliary drainage. A cumulative risk of bleeding after PFCD is reportedly 7%. This may occur because of pseudoaneurysm formation, leading to severe bleeding and hemorrhagic shock (215). Bleeding events have been associated with an increased dwell time of greater than 4 weeks, presumably related to rapid decompression of the cyst and injury to the contralateral wall by the prosthesis (225).

For EUS-GE and EDGE, intraprocedural or postprocedural bleeding can occur for several reasons. The site of bleeding can be intraluminal, intramural, or intraperitoneal (158,159,223). Bleeding is most commonly due to fistula creation or balloon dilation after stent deployment (158,159,170). Other causes include ulceration or erosion by friction from the LAMS (226). Intraprocedural bleeding can be treated by balloon dilation or placement of a larger diameter through-the-scope stent to achieve bleeding tamponade. If this is unsuccessful, then emergent angiography and embolization are warranted. Surgery is rarely needed.

Perforation Bowel perforation after interventional EUS is generally related to stent misdeployment. Less than 1% of EUS-BD cases are complicated by perforation (138,212). However, capnoperitoneum can occur after EUS-BD and EUS-GBD despite the absence of bowel perforation because of gas escape during puncture, sheer forces during dilation, and stent placement. Thus, minimal dilating force and the use of carbon dioxide should be applied (130). Bowel perforation or leakage with peritonitis associated with stent misdeployment has been reported as the most common AE after EUS-GE and EDGE (167,213,214,226,227).

Stent misdeployment Understanding of the stent deployment system is crucial. Misdeployment during EUS-GE can be classified into 4 types (223). Type I, accounting for approximately 63% of events, is defined as the deployment of the distal flange in the peritoneum and proximal flange in the stomach without evidence of a resulting enterotomy. Type II (30.4%) is similar to type I but with an enterotomy (the stent likely migrated out of the small bowel during deployment). Type III (2.2%) is deployment into the small bowel but with the proximal flange in the peritoneum. Type IV (4.4%) is defined as the deployment of the distal flange in the colon resulting in a gastrocolic anastomosis. Early recognition is key as is immediate endoscopic management. In a pure type I injury, stent removal and closure is the preferred technique. This is often also successful in type II misdeployment, as there is ample evidence that small enterotomies may close on their own with no further intervention, provided the patient remains stable. In cases where the patient becomes unstable and complete closure is not possible, a surgical consult should be undertaken (228). Having the patient remain fasted is recommended for at least 24 hours after an AE to ensure proper closure and minimize leakage.

Stent misdeployment is less of a problem for EUS-GBD and EUS-PFCD with the use of a single-step cautery-tip LAMS. However, preloading the cautery-tip LAMS with a guidewire can be applied to ensure safety, and the guidewire can be advanced into the gallbladder or fluid collection before the stent is deployed to secure the puncture tract (229). In case of stent misdeployment, the presence of the guidewire will allow a rescue procedure to be carried out properly. For EUS-BD with metal stents, misdeployment is not unusual. Intrascope channel deployment is recommended to avoid this problem as previously described.

Bile leak Bile leak is the most common AE in EUS-BD, accounting for 3.5%–4.9%, because of the multistep nature of the procedure. Biloma is reported in 0.6% of cases. Bile leakage is inevitable during needle or catheter exchange, guidewire manipulation, and tract dilation; however, this is often clinically unimportant (138,145,148,212,216–219). Using fully covered or partially covered self-expandable metal stents will prevent further bile leakage after placing the stent.

Infection Infection after EUS-PFCD and EUS-GBD is usually related to LAMS occlusion or migration, occasionally requiring urgent endoscopic reintervention to prevent or treat sepsis because of secondary cyst infection or recurrent acute cholecystitis (230,231). The most common infection after EUS-BD is acute cholangitis (2.4%), followed by peritonitis (1.3%) (138,212). As above, prophylactic antibiotics are advisable before performing interventional EUS.

Long-term AEs Long-term or delayed AEs are generally related to stent malfunction. Stent migration, stent occlusion, and buried stent syndrome resulting in delayed bleeding have been reported after EUS-PFCD (215,230–232). Also, delayed bleeding after EUS-BD from cautery-related vascular injury resulting in arteriobiliary fistula has been reported (0.1%) (138). Recurrent acute cholecystitis requiring reintervention after EUS-GBD is reported in 2.6% of cases. However, this event appears to occur less often than after PT-GBD and ERCP-GBD (133,134,137). Indeed, EUS-GBD was associated with a lower cumulative AE at 1 year compared with PT-GBD (77.5% vs 25.6%, $P < 0.001$) (137). Long-term AEs have been reported to occur in about 10% of cases after EUS-GE (214,227). These include stent migration (233), obstruction by food residue, and tissue ingrowth or overgrowth (228), leading to either recurrent symptoms (GOO) or infection, requiring repeat procedures.

Finally, persistent fistula, undermining the gastric bypass, remains a concern after the EDGE procedure (172). This is important because fistulas can lead to weight regain. A LAMS dwell time of 40 days or more was found to have an odds ratio of 4.5 of developing a persistent fistula (odds increased by 9.5% for every 7 days the LAMS was left *in situ*) (172,234).

CONCLUSIONS

This document on quality indicators for EUS mirrors its landscape; most quality indicators are based on the foundations of EUS, but the expanding interventional reach of EUS has necessitated new quality indicators backed by scientific rigor yet tempered when expert opinion supersedes evidence. EUS requires the same prerequisites as any other endoscopic procedure such as a proper informed consent and a well-documented indication including a thorough discussion of potential AEs. In addition, there is a cognitive element that requires image interpretation and delineation of anatomy that are unique to EUS. Detecting and visualizing a lesion might be even more challenging than targeting the lesion or performing an intervention. Recognizing these limitations and proposing alternatives, often using a multidisciplinary team, are key to achieving best outcomes and achieving the quality metrics proposed by this task group. The quality indicators in this iteration balance foundational goals with more aspirational targets, especially in this new era of interventional EUS. The group put great thought in before setting high performance targets, recognizing that success rates and AEs published in the literature are from selected centers with great experience and volumes such that the goals set forth as one chronicles their own EUS journey are achievable. We all recognize that the expansion and adoption of EUS for indications that now seem like a "no-brainer" were unimaginable even 5 years ago. With this evolution, the ACG and ASGE plan to continually reassess this evidence and modify quality indicators for EUS accordingly.

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CONFLICTS OF INTEREST

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REFERENCES

1. Schulman AR, Howell JD. From Hirschowitz to 2023: Modern endoscopy and beyond. Clin Gastroenterol Hepatol 2024;22(4):684–8.
2. Chassin MR, Galvin RW. The urgent need to improve health care quality. Institute of Medicine National Roundtable on Health Care Quality. JAMA 1998;280(11):1000–5.
3. Jacobson BC, Chak A, Hoffman B, et al. Quality indicators for endoscopic ultrasonography. Am J Gastroenterol 2006;101(4):898–901.
4. Wani S, Wallace MB, Cohen J, et al. Quality indicators for EUS. Gastrointest Endosc 2015;81(1):67–80.
5. Elmunzer BJ, Anderson MA, Mishra G, et al. Quality indicators common to all gastrointestinal endoscopic procedures. Am J Gastroenterol 2024;119(9):1781–91.
6. Day LW, Cohen J, Greenwald D, et al. Quality indicators for gastrointestinal endoscopy units. VideoGIE 2017;2(6):119–40.
7. Li S, Monachese M, Salim M, et al. Standard reporting elements for the performance of EUS: Recommendations from the FOCUS working group. Endosc Ultrasound 2021;10(2):84–92.
8. Shami VM, Mahajan A, Loch MM, et al. Comparison between endoscopic ultrasound and magnetic resonance imaging for the staging of pancreatic cancer. Pancreas 2011;40(4):567–70.
9. Pietryga JA, Morgan DE. Imaging preoperatively for pancreatic adenocarcinoma. J Gastrointest Oncol 2015;6(4):343–57.
10. Singh A, Faulx AL. Endoscopic evaluation in the workup of pancreatic cancer. Surg Clin North Am 2016;96(6):1257–70.
11. Aslanian HR, Lee JH, Canto MI. AGA clinical practice update on pancreas cancer screening in high-risk individuals: Expert review. Gastroenterology 2020;159(1):358–62.
12. Sawhney MS, Calderwood AH, Thosani NC, et al. ASGE guideline on screening for pancreatic cancer in individuals with genetic susceptibility: Summary and recommendations. Gastrointest Endosc 2022;95(5): 817–26.
13. Ryou M, DeWitt JM, Das KK, et al. AGA clinical practice update on interventional EUS for vascular investigation and therapy: Commentary. Clin Gastroenterol Hepatol 2023;21(7):1699–705.e2.
14. Khan Z, Hayat U, Moraveji S, et al. EUS-guided pancreatic ductal intervention: A comprehensive literature review. Endosc Ultrasound 2021;10(2):98–102.
15. Nakai Y, Isayama H, Yamamoto N, et al. Indications for endoscopic ultrasonography (EUS)-guided biliary intervention: Does EUS always come after failed endoscopic retrograde cholangiopancreatography? Dig Endosc 2017;29(2):218–25.
16. Prager M, Prager E, Sebesta C Jr, et al. Diagnostic and therapeutic indications for endoscopic ultrasound (EUS) in patients with pancreatic and biliary disease: Novel interventional procedures. Curr Oncol 2022; 29(9):6211–25.
17. Jearth V, Sundaram S, Rana SS. Diagnostic and interventional EUS in hepatology: An updated review. Endosc Ultrasound 2022;11(5):355–70.
18. Abdelqader A, Kahaleh M. When ERCP fails: EUS-guided access to biliary and pancreatic ducts. Dig Dis Sci 2022;67(5):1649–59.
19. Forbes N, Coelho-Prabhu N, Al-Haddad MA, et al. Adverse events associated with EUS and EUS-guided procedures. Gastrointest Endosc 2022;95(1):16–26.e2.
20. Colán-Hernández J, Sendino O, Loras C, et al. Antibiotic prophylaxis is not required for endoscopic ultrasonography-guided fine-needle aspiration of pancreatic cystic lesions, based on a randomized trial. Gastroenterology 2020;158(6):1642–9.e1.
21. Palomera-Tejeda E, Shah H, Attar BM, et al. Prophylactic antibiotics do not prevent infectious complications of endoscopic ultrasound fine-needle aspiration of pancreatic cysts: A systematic review and meta-analysis. Pancreas 2021;50(5):667–72.
22. Facciorusso A, Mohan BP, Tacelli M, et al. Use of antibiotic prophylaxis is not needed for endoscopic ultrasound-guided fine-needle aspiration of pancreatic cysts: A meta-analysis. Expert Rev Gastroenterol Hepatol 2020;14(10):999–1005.
23. Facciorusso A, Arevalo-Mora M, Conti Bellocchi MC, et al. Impact of antibiotic prophylaxis on infection rate after endoscopic ultrasound through-the-needle biopsy of pancreatic cysts: A propensity score-matched study. Diagnostics 2022;12(1):211.
24. Facciorusso A, Buccino VR, Turco A, et al. Antibiotics do not decrease the rate of infection after endoscopic ultrasound fine-needle aspiration of pancreatic cysts. Dig Dis Sci 2019;64(8):2308–15.
25. Ashhab AA, Pandol SJ, Lo SK. Are we ready to give up prophylactic antibiotics for endoscopic ultrasonography-guided fine-needle aspiration of pancreatic cystic lesions? Am J Gastroenterol 2021;116(12):2482–3.
26. Kaushik N, Khalid A, Brody D, et al. EUS-guided paracentesis for the diagnosis of malignant ascites. Gastrointest Endosc 2006;64(6):908–13.
27. Valli PV, Gubler C, Bauerfeind P. Severe infectious complications after endoscopic ultrasound-guided fine needle aspiration of suspected mediastinal duplication cysts: A case series. Inflamm Intest Dis 2017;1(4):165–71.
28. Soh JS, Lee HS, Lee S, et al. The clinical usefulness of endoscopic ultrasound-guided fine needle aspiration and biopsy for rectal and perirectal lesions. Intest Res 2015;13(2):135–44.
29. Brown LK, Clark NR, Conway J, et al. Rectal ultrasound with fine needle aspiration: An underutilized modality for delineating and diagnosing perirectal, presacral, and pelvic lesions. Endosc Int Open 2019;7(2): E171–7.
30. van Wanrooij RLJ, Bronswijk M, Kunda R, et al. Therapeutic endoscopic ultrasound: European Society of Gastrointestinal Endoscopy (ESGE) technical review. Endoscopy 2022;54(3):310–32.
31. Moon E, Tam MD, Kikano RN, et al. Prophylactic antibiotic guidelines in modern interventional radiology practice. Semin Intervent Radiol 2010;27(4):327–37.
32. Berríos-Torres SI, Umscheid CA, Bratzler DW, et al. Centers for Disease Control and Prevention guideline for the prevention of surgical site infection, 2017. JAMA Surg 2017;152(8):784–91.
33. Qayed E, Sheth SG, Aihara H, et al. Advanced endoscopy fellowship training in the United States: Recent trends in American Society for Gastrointestinal Endoscopy advanced endoscopy fellowship match, trainee experience, and postfellowship employment. Gastrointest Endosc 2021;93(6):1207–14.e2.
34. Yang D, Wagh MS, Draganov PV. The status of training in new technologies in advanced endoscopy: From defining competence to credentialing and privileging. Gastrointest Endosc 2020;92(5):1016–25.
35. Götzberger M, Kaiser C, Jenssen C, et al. A structured endoscopic ultrasound training program in Germany improves knowledge and competence. Med Ultrason 2021;23(4):396–401.
36. Cassani L, Aihara H, Anand GS, et al. Core curriculum for EUS. Gastrointest Endosc 2020;92(3):469–73.
37. Karstensen JG, Nayahangan LJ, Konge L, et al. A core curriculum for basic EUS skills: An international consensus using the Delphi methodology. Endosc Ultrasound 2022;11(2):122–32.

38. Johnson G, Webster G, Boškoski I, et al. Curriculum for ERCP and endoscopic ultrasound training in Europe: European Society of Gastrointestinal Endoscopy (ESGE) position statement. Endoscopy 2021;53(10):1071–87.
39. Hedenström P, Marasco G, Eusebi LH, et al. GAPS-EUS: A new and reliable tool for the assessment of basic skills and performance in EUS among endosonography trainees. BMJ Open Gastroenterol 2021;8(1): e000660.
40. Wani S, Han S, Simon V, et al. Setting minimum standards for training in EUS and ERCP: Results from a prospective multicenter study evaluating learning curves and competence among advanced endoscopy trainees. Gastrointest Endosc 2019;89(6):1160–8.e9.
41. Faulx AL, Lightdale JR, Acosta RD, et al. Guidelines for privileging, credentialing, and proctoring to perform GI endoscopy. Gastrointest Endosc 2017;85(2):273–81.
42. Worrell SG, Oh DS, Greene CL, et al. Endoscopic ultrasound staging of stenotic esophageal cancers may be unnecessary to determine the need for neoadjuvant therapy. J Gastrointest Surg 2014;18(2):318–20.
43. Rice TW, Gress DM, Patil DT, et al. Cancer of the esophagus and esophagogastric junction: Major changes in the American Joint Committee on Cancer eighth edition cancer staging manual. CA Cancer J Clin 2017;67(4):304–17.
44. Cen P, Hofstetter WL, Lee JH, et al. Value of endoscopic ultrasound staging in conjunction with the evaluation of lymphovascular invasion in identifying low-risk esophageal carcinoma. Cancer 2008;112(3): 503–10.
45. Krill T, Baliss M, Roark R, et al. Accuracy of endoscopic ultrasound in esophageal cancer staging. J Thorac Dis 2019;11(Suppl 12):S1602–9.
46. Reddy RP, Levy MJ, Wiersema MJ. Endoscopic ultrasound for luminal malignancies. Gastrointest Endosc Clin North Am 2005;15(3):399–429, vii.
47. Tamerisa R, Irisawa A, Bhutani MS. Endoscopic ultrasound in the diagnosis, staging, and management of gastrointestinal and adjacent malignancies. Med Clin North Am 2005;89(1):139–58, viii.
48. Foley KG, Christian A, Fielding P, et al. Accuracy of contemporary oesophageal cancer lymph node staging with radiological-pathological correlation. Clin Radiol 2017;72(8):693.e1–7.
49. Fairweather M, Jajoo K, Sainani N, et al. Accuracy of EUS and CT imaging in preoperative gastric cancer staging. J Surg Oncol 2015;111(8): 1016–20.
50. Lee KG, Shin CI, Kim SG, et al. Can endoscopic ultrasonography (EUS) improve the accuracy of clinical T staging by computed tomography (CT) for gastric cancer? Eur J Surg Oncol 2021;47(8):1969–75.
51. Hünerbein M. Endorectal ultrasound in rectal cancer. Colorectal Dis 2003;5:402–5.
52. Campbell C, Conway J, Lavette LE, et al. Endorectal ultrasound (ERUS): An accurate and invaluable tool for identifying suitable candidates for transanal excision/transanal endoscopic microsurgery (TAE/TEM) in early-staged rectal tumors. J Gastrointest Surg 2023; 27(12):3086–7.
53. Fernandes MC, Gollub MJ, Brown G. The importance of MRI for rectal cancer evaluation. Surg Oncol 2022;43:101739.
54. DeMeester SR. Evaluation and treatment of superficial esophageal cancer. J Gastrointest Surg 2010;14(Suppl 1):S94–100.
55. Nobel T, Sihag S. Advances in diagnostic, staging, and restaging evaluation of esophageal and gastric cancer. Surg Oncol Clin North Am 2024;33(3):467–85.
56. Ajani JA, D'Amico TA, Bentrem DJ, et al. Esophageal and esophagogastric junction cancers, version 2.2023, NCCN clinical practice guidelines in oncology. J Natl Compr Cancer Netw 2023;21(4): 393–422.
57. Gong EJ, Kim DH. Endoscopic ultrasonography in the diagnosis of gastric subepithelial lesions. Clin Endosc 2016;49(5):425–33.
58. Rösch T, Kapfer B, Will U, et al. Accuracy of endoscopic ultrasonography in upper gastrointestinal submucosal lesions: A prospective multicenter study. Scand J Gastroenterol 2002;37(7):856–62.
59. Motoo Y, Okai T, Ohta H, et al. Endoscopic ultrasonography in the diagnosis of extraluminal compressions mimicking gastric submucosal tumors. Endoscopy 1994;26(2):239–42.
60. Hwang JH, Saunders MD, Rulyak SJ, et al. A prospective study comparing endoscopy and EUS in the evaluation of GI subepithelial masses. Gastrointest Endosc 2005;62(2):202–8.
61. Kim SY, Shim KN, Lee JH, et al. Comparison of the diagnostic ability of endoscopic ultrasonography and abdominopelvic computed tomography in the diagnosis of gastric subepithelial tumors. Clin Endosc 2019;52(6):565–73.
62. Seifert H, Fusaroli P, Arcidiacono PG, et al. Controversies in EUS: Do we need miniprobes? Endosc Ultrasound 2021;10(4):246–69.
63. Xu GQ, Li YW, Han YM, et al. Miniature ultrasonic probes for diagnosis and treatment of digestive tract diseases. World J Gastroenterol 2004; 10(13):1948–53.
64. Khan S, Zhang R, Fang W, et al. Reliability of endoscopic ultrasound using miniprobes and grayscale histogram analysis in diagnosing upper gastrointestinal subepithelial lesions. Gastroenterol Res Pract 2020; 2020:65913416591341.
65. Karaca C, Turner BG, Cizginer S, et al. Accuracy of EUS in the evaluation of small gastric subepithelial lesions. Gastrointest Endosc 2010;71(4):722–7.
66. Reddymasu SC, Oropeza-Vail M, Pakseresht K, et al. Are endoscopic ultrasonography imaging characteristics reliable for the diagnosis of small upper gastrointestinal subepithelial lesions? J Clin Gastroenterol 2012;46(1):42–5.
67. Brand B, Oesterhelweg L, Binmoeller KF, et al. Impact of endoscopic ultrasound for evaluation of submucosal lesions in gastrointestinal tract. Dig Liver Dis 2002;34(4):290–7.
68. de Moura DTH, McCarty TR, Jirapinyo P, et al. EUS-guided fine-needle biopsy sampling versus FNA in the diagnosis of subepithelial lesions: A large multicenter study. Gastrointest Endosc 2020;92(1):108–19.e3.
69. Sharzehi K, Sethi A, Savides T. AGA clinical practice update on management of subepithelial lesions encountered during routine endoscopy: Expert review. Clin Gastroenterol Hepatol 2022;20(11):2435–43.e4.
70. Jacobson BC, Bhatt A, Greer KB, et al. ACG clinical guideline: Diagnosis and management of gastrointestinal subepithelial lesions. Am J Gastroenterol 2023;118(1):46–58.
71. Facciorusso A, Sunny SP, Del Prete V, et al. Comparison between fine-needle biopsy and fine-needle aspiration for EUS-guided sampling of subepithelial lesions: A meta-analysis. Gastrointest Endosc 2020;91(1): 14–22.e2.
72. Zhang XC, Li QL, Yu YF, et al. Diagnostic efficacy of endoscopic ultrasound-guided needle sampling for upper gastrointestinal subepithelial lesions: A meta-analysis. Surg Endosc 2016;30(6):2431–41.
73. Rockey DC, Caldwell SH, Goodman ZD, et al. Liver biopsy. Hepatology 2009;49(3):1017–44.
74. Fryer E, Wang LM, Verrill C, et al. How often do our liver core biopsies reach current definitions of adequacy? J Clin Pathol 2013;66(12):1087–9.
75. Neuberger J, Patel J, Caldwell H, et al. Guidelines on the use of liver biopsy in clinical practice from the British Society of Gastroenterology, the Royal College of Radiologists and the Royal College of Pathology. Gut 2020;69(8):1382–403.
76. Mohan BP, Shakhatreh M, Garg R, et al. Efficacy and safety of EUS-guided liver biopsy: A systematic review and meta-analysis. Gastrointest Endosc 2019;89(2):238–46.e3.
77. Zeng K, Jiang Z, Yang J, et al. Role of endoscopic ultrasound-guided liver biopsy: A meta-analysis. Scand J Gastroenterol 2022;57(5):545–57.
78. Baran B, Kale S, Patil P, et al. Endoscopic ultrasound-guided parenchymal liver biopsy: A systematic review and meta-analysis. Surg Endosc 2021;35(10):5546–57.
79. Chandan S, Deliwala S, Khan SR, et al. EUS-guided versus percutaneous liver biopsy: A comprehensive review and meta-analysis of outcomes. Endosc Ultrasound 2023;12(2):171–80.
80. McCarty TR, Bazarbashi AN, Njei B, et al. Endoscopic ultrasound-guided, percutaneous, and transjugular liver biopsy: A comparative systematic review and meta-analysis. Clin Endosc 2020;53(5):583–93.
81. Facciorusso A, Crinò SF, Ramai D, et al. Diagnostic yield of endoscopic ultrasound-guided liver biopsy in comparison to percutaneous liver biopsy: A systematic review and meta-analysis. Expert Rev Gastroenterol Hepatol 2022;16(1):51–7.
82. Bang JY, Ward TJ, Guirguis S, et al. Radiology-guided percutaneous approach is superior to EUS for performing liver biopsies. Gut 2021; 70(12):2224–6.
83. Lariño-Noia J, Fernández-Castroagudín J, de la Iglesia-García D, et al. Quality of tissue samples obtained by endoscopic ultrasound-guided liver biopsy: A randomized, controlled clinical trial. Am J Gastroenterol 2023;118(10):1821–8.
84. Hashimoto R, Lee DP, Samarasena JB, et al. Comparison of two specialized histology needles for endoscopic ultrasound (EUS)-guided liver biopsy: A pilot study. Dig Dis Sci 2021;66(5):1700–6.
85. Nieto J, Dawod E, Deshmukh A, et al. EUS-guided fine-needle core liver biopsy with a modified one-pass, one-actuation wet suction technique

comparing two types of EUS core needles. Endosc Int Open 2020;8(7): E938–43.
86. Zakaria A, Diab A, Dowd M, et al. EUS-guided liver biopsy using a modified wet heparin suction technique. VideoGIE 2022;7(10):348–9.
87. Mok SRS, Diehl DL, Johal AS, et al. A prospective pilot comparison of wet and dry heparinized suction for EUS-guided liver biopsy (with videos). Gastrointest Endosc 2018;88(6):919–25.
88. Diehl DL, Sangwan V, Johal AS, et al. Comparing a 19-gauge fine-needle biopsy needle with a 22-gauge fine-needle biopsy needle for EUS-guided liver biopsy sampling: A prospective randomized study. Gastrointest Endosc 2024;99(6):931–7.
89. Best LM, Rawji V, Pereira SP, et al. Imaging modalities for characterising focal pancreatic lesions. Cochrane Database Syst Rev 2017;4:CD010213.
90. Dewitt J, Devereaux BM, Lehman GA, et al. Comparison of endoscopic ultrasound and computed tomography for the preoperative evaluation of pancreatic cancer: A systematic review. Clin Gastroenterol Hepatol 2006;4(6):717–25; quiz 664.
91. Krishna SG, Rao BB, Ugbarugba E, et al. Diagnostic performance of endoscopic ultrasound for detection of pancreatic malignancy following an indeterminate multidetector CT scan: A systemic review and meta-analysis. Surg Endosc 2017;31(11):4558–67.
92. Kitano M, Kudo M, Yamao K, et al. Characterization of small solid tumors in the pancreas: The value of contrast-enhanced harmonic endoscopic ultrasonography. Am J Gastroenterol 2012;107(2):303–10.
93. Li Y, Jin H, Liao D, et al. Contrast-enhanced harmonic endoscopic ultrasonography for the differential diagnosis of pancreatic masses: A systematic review and meta-analysis. Mol Clin Oncol 2019;11(4): 425–33.
94. Gheonea DI, Săftoiu A. Beyond conventional endoscopic ultrasound: Elastography, contrast enhancement and hybrid techniques. Curr Opin Gastroenterol 2011;27(5):423–9.
95. Hirooka Y, Itoh A, Kawashima H, et al. Diagnosis of pancreatic disorders using contrast-enhanced endoscopic ultrasonography and endoscopic elastography. Clin Gastroenterol Hepatol 2009;7(11 Suppl):S63–7.
96. Abdallah MA, Ahmed K, Taha W, et al. Endoscopic ultrasound guided fine-needle aspiration for solid lesions in chronic pancreatitis: A systematic review and meta-analysis. Dig Dis Sci 2022;67(6):2552–61.
97. Varadarajulu S, Tamhane A, Eloubeidi MA. Yield of EUS-guided FNA of pancreatic masses in the presence or the absence of chronic pancreatitis. Gastrointest Endosc 2005;62(5):728–36; quiz 751, 753.
98. Chun YS, Pawlik TM, Vauthey JN. 8th Edition of the AJCC cancer staging manual: Pancreas and hepatobiliary cancers. Ann Surg Oncol 2018;25(4):845–7.
99. Nawaz H, Fan CY, Kloke J, et al. Performance characteristics of endoscopic ultrasound in the staging of pancreatic cancer: A meta-analysis. J Pancreas 2013;14(5):484–97.
100. Li AE, Li BT, Ng BHK, et al. Diagnostic accuracy of imaging modalities in the evaluation of vascular invasion in pancreatic adenocarcinoma: A meta-analysis. World J Oncol 2013;4(2):74–82.
101. DeWitt J, LeBlanc J, McHenry L, et al. Endoscopic ultrasound-guided fine-needle aspiration of ascites. Clin Gastroenterol Hepatol 2007;5: 609–15.
102. Minaga K, Kitano M, Nakai A, et al. Improved detection of liver metastasis using Kupffer-phase imaging in contrast-enhanced harmonic EUS in patients with pancreatic cancer (with video). Gastrointest Endosc 2021;93(2):433–41.
103. Okasha HH, Wifi MN, Awad A, et al. Role of EUS in detection of liver metastasis not seen by computed tomography or magnetic resonance imaging during staging of pancreatic, gastrointestinal, and thoracic malignancies. Endosc Ultrasound 2021;10(5):344–54.
104. Hewitt MJ, McPhail MJ, Possamai L, et al. EUS-guided FNA for diagnosis of solid pancreatic neoplasms: A meta-analysis. Gastrointest Endosc 2012;75(2):319–31.
105. Puli SR, Bechtold ML, Buxbaum JL, et al. How good is endoscopic ultrasound-guided fine-needle aspiration in diagnosing the correct etiology for a solid pancreatic mass? A meta-analysis and systematic review. Pancreas 2013;42(1):20–6.
106. Li H, Li W, Zhou QY, et al. Fine needle biopsy is superior to fine needle aspiration in endoscopic ultrasound guided sampling of pancreatic masses: A meta-analysis of randomized controlled trials. Medicine 2018; 97(13):e0207.
107. Facciorusso A, Wani S, Triantafyllou K, et al. Comparative accuracy of needle sizes and designs for EUS tissue sampling of solid pancreatic masses: A network meta-analysis. Gastrointest Endosc 2019;90(6):893–903.e7.
108. van Riet PA, Erler NS, Bruno MJ, et al. Comparison of fine-needle aspiration and fine-needle biopsy devices for endoscopic ultrasound-guided sampling of solid lesions: A systemic review and meta-analysis. Endoscopy 2021;53(4):411–23.
109. Gkolfakis P, Crinò SF, Tziatzios G, et al. Comparative diagnostic performance of end-cutting fine-needle biopsy needles for EUS tissue sampling of solid pancreatic masses: A network meta-analysis. Gastrointest Endosc 2022;95(6):1067–77.e15.
110. Han S, Bhullar F, Alaber O, et al. Comparative diagnostic accuracy of EUS needles in solid pancreatic masses: A network meta-analysis. Endosc Int Open 2021;9(6):E853–62.
111. Affolter KE, Schmidt RL, Matynia AP, et al. Needle size has only a limited effect on outcomes in EUS-guided fine needle aspiration: A systematic review and meta-analysis. Dig Dis Sci 2013;58(4):1026–34.
112. Guedes HG, Moura DTH, Duarte RB, et al. A comparison of the efficiency of 22G versus 25G needles in EUS-FNA for solid pancreatic mass assessment: A systematic review and meta-analysis. Clinics 2018; 73:e261.
113. Madhoun MF, Wani SB, Rastogi A, et al. The diagnostic accuracy of 22-gauge and 25-gauge needles in endoscopic ultrasound-guided fine needle aspiration of solid pancreatic lesions: A meta-analysis. Endoscopy 2013;45(2):86–92.
114. Xu MM, Jia HY, Yan LL, et al. Comparison of two different size needles in endoscopic ultrasound-guided fine-needle aspiration for diagnosing solid pancreatic lesions: A meta-analysis of prospective controlled trials. Medicine 2017;96(5):e5802.
115. Parks RW, Tzovaras G, Diamond T, et al. Management of pancreatic pseudocysts. Ann R Coll Surg Engl 2000;82(6):383–7.
116. Johnson MD, Walsh RM, Henderson JM, et al. Surgical versus nonsurgical management of pancreatic pseudocysts. J Clin Gastroenterol 2009;43(6):586–90.
117. Varadarajulu S, Bang JY, Sutton BS, et al. Equal efficacy of endoscopic and surgical cystogastrostomy for pancreatic pseudocyst drainage in a randomized trial. Gastroenterology 2013;145(3):583–90.e1.
118. Park DH, Lee SS, Moon SH, et al. Endoscopic ultrasound-guided versus conventional transmural drainage for pancreatic pseudocysts: A prospective randomized trial. Endoscopy 2009;41(10): 842–8.
119. Siddiqui AA, Kowalski TE, Loren DE, et al. Fully covered self-expanding metal stents versus lumen-apposing fully covered self-expanding metal stent versus plastic stents for endoscopic drainage of pancreatic walled-off necrosis: Clinical outcomes and success. Gastrointest Endosc 2017; 85(4):758–65.
120. Bang JY, Navaneethan U, Hasan MK, et al. Non-superiority of lumen-apposing metal stents over plastic stents for drainage of walled-off necrosis in a randomised trial. Gut 2019;68(7):1200–9.
121. Patil R, Ona MA, Papafragkakis C, et al. Endoscopic ultrasound-guided placement of AXIOS stent for drainage of pancreatic fluid collections. Ann Gastroenterol 2016;29(2):168–73.
122. Shah RJ, Shah JN, Waxman I, et al. Safety and efficacy of endoscopic ultrasound-guided drainage of pancreatic fluid collections with lumen-apposing covered self-expanding metal stents. Clin Gastroenterol Hepatol 2015;13(4):747–52.
123. van Brunschot S, van Grinsven J, van Santvoort HC, et al. Endoscopic or surgical step-up approach for infected necrotising pancreatitis: A multicentre randomised trial. Lancet 2018;391(10115):51–8.
124. van Santvoort HC, Besselink MG, Bakker OJ, et al. A step-up approach or open necrosectomy for necrotizing pancreatitis. N Engl J Med 2010; 362(16):1491–502.
125. Boxhoorn L, van Dijk SM, van Grinsven J, et al. Immediate versus postponed intervention for infected necrotizing pancreatitis. N Engl J Med 2021;385(15):1372–81.
126. Kumta NA, Tyberg A, Bhagat VH, et al. EUS-guided drainage of pancreatic fluid collections using lumen apposing metal stents: An international, multicenter experience. Dig Liver Dis 2019;51(11): 1557–61.
127. Karstensen JG, Novovic S, Hansen EF, et al. EUS-guided drainage of large walled-off pancreatic necroses using plastic versus lumen-apposing metal stents: A single-centre randomised controlled trial. Gut 2023; 72(6):1167–73.
128. Saumoy M, Tyberg A, Brown E, et al. Successful cholecystectomy after endoscopic ultrasound gallbladder drainage compared with percutaneous cholecystostomy: Can it be done? J Clin Gastroenterol 2019;53(3):231–5.

129. Irani SS, Sharzehi K, Siddiqui UD. AGA clinical practice update on role of EUS-guided gallbladder drainage in acute cholecystitis: Commentary. Clin Gastroenterol Hepatol 2023;21(5):1141–7.
130. Rana SS. Endoscopic ultrasound-guided gallbladder drainage: A technical review. Ann Gastroenterol 2021;34(2):142–8.
131. Robles-Medranda C, Oleas R, Puga-Tejada M, et al. Prophylactic EUS-guided gallbladder drainage prevents acute cholecystitis in patients with malignant biliary obstruction and cystic duct orifice involvement: A randomized trial (with video). Gastrointest Endosc 2023;97(3):445–53.
132. Tyberg A, Duarte-Chavez R, Shahid HM, et al. Endoscopic ultrasound-guided gallbladder drainage versus percutaneous drainage in patients with acute cholecystitis undergoing elective cholecystectomy. Clin Transl Gastroenterol 2023;14(6):e00593.
133. Mohan BP, Khan SR, Trakroo S, et al. Endoscopic ultrasound-guided gallbladder drainage, transpapillary drainage, or percutaneous drainage in high risk acute cholecystitis patients: A systematic review and comparative meta-analysis. Endoscopy 2020;52(2):96–106.
134. Podboy A, Yuan J, Stave CD, et al. Comparison of EUS-guided endoscopic transpapillary and percutaneous gallbladder drainage for acute cholecystitis: A systematic review with network meta-analysis. Gastrointest Endosc 2021;93(4):797–804.e1.
135. Jang JW, Lee SS, Song TJ, et al. Endoscopic ultrasound-guided transmural and percutaneous transhepatic gallbladder drainage are comparable for acute cholecystitis. Gastroenterology 2012;142(4): 805–11.
136. Krishnamoorthi R, Jayaraj M, Thoguluva Chandrasekar V, et al. EUS-guided versus endoscopic transpapillary gallbladder drainage in high-risk surgical patients with acute cholecystitis: A systematic review and meta-analysis. Surg Endosc 2020;34(5):1904–13.
137. Teoh AYB, Kitano M, Itoi T, et al. Endosonography-guided gallbladder drainage versus percutaneous cholecystostomy in very high-risk surgical patients with acute cholecystitis: An international randomised multicentre controlled superiority trial (DRAC 1). Gut 2020;69(6): 1085–91.
138. Dhir V, Shah R, Udawat P. Endoscopic ultrasound-guided biliary interventions. Gastrointest Endosc Clin North Am 2022;32(3):507–25.
139. Elmunzer BJ, Maranki JL, Gómez V, et al. ACG clinical guideline: Diagnosis and management of biliary strictures. Am J Gastroenterol 2023;118(3):405–26.
140. Chen YI, Sahai A, Donatelli G, et al. Endoscopic ultrasound-guided biliary drainage of first intent with a lumen-apposing metal stent vs endoscopic retrograde cholangiopancreatography in malignant distal biliary obstruction: A multicenter randomized controlled study (ELEMENT Trial). Gastroenterology 2023;165(5):1249–61.e5.
141. Teoh AYB, Napoleon B, Kunda R, et al. EUS-guided choledocho-duodenostomy using lumen apposing stent versus ERCP with covered metallic stents in patients with unresectable malignant distal biliary obstruction: A multicenter randomized controlled trial (DRA-MBO Trial). Gastroenterology 2023;165(2):473–82.e2.
142. Canakis A, Baron TH. Therapeutic endoscopic ultrasound: Current indications and future perspectives. GE Port J Gastroenterol 2023; 30(Suppl 1):4–18.
143. Canakis A, Baron TH. Relief of biliary obstruction: Choosing between endoscopic ultrasound and endoscopic retrograde cholangiopancreatography. BMJ Open Gastroenterol 2020;7(1): e000428.
144. Kawakubo K, Isayama H, Kato H, et al. Multicenter retrospective study of endoscopic ultrasound-guided biliary drainage for malignant biliary obstruction in Japan. J Hepatobiliary Pancreat Sci 2014;21(5):328–34.
145. Park DH, Jang JW, Lee SS, et al. EUS-guided biliary drainage with transluminal stenting after failed ERCP: Predictors of adverse events and long-term results. Gastrointest Endosc 2011;74(6):1276–84.
146. Bang JY, Navaneethan U, Hasan M, et al. Stent placement by EUS or ERCP for primary biliary decompression in pancreatic cancer: A randomized trial (with videos). Gastrointest Endosc 2018;88(1):9–17.
147. Paik WH, Lee TH, Park DH, et al. EUS-guided biliary drainage versus ERCP for the primary palliation of malignant biliary obstruction: A multicenter randomized clinical trial. Am J Gastroenterol 2018;113(7):987–97.
148. Dhir V, Itoi T, Khashab MA, et al. Multicenter comparative evaluation of endoscopic placement of expandable metal stents for malignant distal common bile duct obstruction by ERCP or EUS-guided approach. Gastrointest Endosc 2015;81(4):913–23.
149. Ogura T, Chiba Y, Masuda D, et al. Comparison of the clinical impact of endoscopic ultrasound-guided choledochoduodenostomy and hepaticogastrostomy for bile duct obstruction with duodenal obstruction. Endoscopy 2016;48(2):163.
150. Ogura T, Higuchi K. Technical review of developments in endoscopic ultrasound-guided hepaticogastrostomy. Clin Endosc 2021;54(5): 651–9.
151. Chantarojanasiri T, Ratanachu-Ek T, Pausawasdi N. What you need to know before performing endoscopic ultrasound-guided hepaticogastrostomy. Clin Endosc 2021;54(3):301–8.
152. Nakamura J, Ogura T, Ueno S, et al. Liver impaction technique improves technical success rate of guidewire insertion during EUS-guided hepaticogastrostomy (with video). Therap Adv Gastroenterol 2023;16: 17562848231188562.
153. Tsuchiya T, Itoi T, Ishii K, et al. Long-term outcomes of EUS-guided balloon-occluded gastrojejunostomy bypass for malignant gastric outlet obstruction (with video). Gastrointest Endosc 2025;101(1): 195–9.
154. Tonozuka R, Tsuchiya T, Mukai S, et al. Endoscopic ultrasonography-guided gastroenterostomy techniques for treatment of malignant gastric outlet obstruction. Clin Endosc 2020;53(5):510–8.
155. Tringali A, Giannetti A, Adler DG. Endoscopic management of gastric outlet obstruction disease. Ann Gastroenterol 2019;32(4):330–7.
156. Bronswijk M, Fransen L, Vanella G, et al. Successful treatment of superior mesenteric artery syndrome by endoscopic ultrasound-guided gastrojejunostomy. Endoscopy 2021;53(2):204–5.
157. De Ponthaud C, Bozkirli B, Rizzo GEM, et al. Management of malignant gastric outlet obstruction (mGOO) due to pancreatic cancer in the era of EUS-gastrojejunostomy: An international practice survey and case vignette study by Pancreas 2000 from the European Pancreatic Club. Surg Endosc 2024;38(6):3231–40.
158. Perez-Miranda M, Tyberg A, Poletto D, et al. EUS-guided gastrojejunostomy versus laparoscopic gastrojejunostomy: An international collaborative study. J Clin Gastroenterol 2017;51(10):896–9.
159. Khashab MA, Bukhari M, Baron TH, et al. International multicenter comparative trial of endoscopic ultrasonography-guided gastroenterostomy versus surgical gastrojejunostomy for the treatment of malignant gastric outlet obstruction. Endosc Int Open 2017;5(4): E275–81.
160. Chen YI, Itoi T, Baron TH, et al. EUS-guided gastroenterostomy is comparable to enteral stenting with fewer re-interventions in malignant gastric outlet obstruction. Surg Endosc 2017;31(7):2946–52.
161. Itoi T, Baron TH, Khashab MA, et al. Technical review of endoscopic ultrasonography-guided gastroenterostomy in 2017. Dig Endosc 2017; 29(4):495–502.
162. Chan SM, Dhir V, Chan YYY, et al. Endoscopic ultrasound-guided balloon-occluded gastrojejunostomy bypass, duodenal stent or laparoscopic gastrojejunostomy for unresectable malignant gastric outlet obstruction. Dig Endosc 2023;35(4):512–9.
163. Canakis A, Bomman S, Lee DU, et al. Benefits of EUS-guided gastroenterostomy over surgical gastrojejunostomy in the palliation of malignant gastric outlet obstruction: A large multicenter experience. Gastrointest Endosc 2023;98(3):348–59.e30.
164. Tyberg A, Perez-Miranda M, Sanchez-Ocaña R, et al. Endoscopic ultrasound-guided gastrojejunostomy with a lumen-apposing metal stent: A multicenter, international experience. Endosc Int Open 2016; 4(3):E276–81.
165. Marrache MK, Itani MI, Farha J, et al. Endoscopic gastrointestinal anastomosis: A review of established techniques. Gastrointest Endosc 2021;93(1):34–46.
166. Li JS, Lin K, Tang J, et al. EUS-guided gastroenterostomy for gastric outlet obstruction: A comprehensive meta-analysis. Minim Invasive Ther Allied Technol 2023;32(6):285–99.
167. Kedia P, Sharaiha RZ, Kumta NA, et al. Internal EUS-directed transgastric ERCP (EDGE): Game over. Gastroenterology 2014;147(3):566–8.
168. Runge TM, Chiang AL, Kowalski TE, et al. Endoscopic ultrasound-directed transgastric ERCP (EDGE): A retrospective multicenter study. Endoscopy 2021;53(6):611–8.
169. Tønnesen CJ, Young J, Glomsaker T, et al. Laparoscopy-assisted versus balloon enteroscopy-assisted ERCP after Roux-en-Y gastric bypass. Endoscopy 2020;52(8):654–61.
170. Ghandour B, Shinn B, Dawod QM, et al. EUS-directed transgastric interventions in Roux-en-Y gastric bypass anatomy: A multicenter experience. Gastrointest Endosc 2022;96(4):630–8.
171. Kedia P, Tarnasky PR, Nieto J, et al. EUS-directed transgastric ERCP (EDGE) versus laparoscopy-assisted ERCP (LA-ERCP) for Roux-en-Y

gastric bypass (RYGB) anatomy: A multicenter early comparative experience of clinical outcomes. J Clin Gastroenterol 2019;53(4):304–8.
172. Ghandour B, Keane MG, Shinn B, et al. Factors predictive of persistent fistulas in EUS-directed transgastric ERCP: A multicenter matched case-control study. Gastrointest Endosc 2023;97(2):260–7.
173. Kedia P, Shah-Khan S, Tyberg A, et al. Endoscopic ultrasound-directed transgastric ERCP (EDGE): A multicenter US study on long-term follow-up and fistula closure. Endosc Int Open 2023;11(5):E529–37.
174. Prakash S, Elmunzer BJ, Forster EM, et al. Endoscopic ultrasound-directed transgastric ERCP (EDGE): A systematic review describing the outcomes, adverse events, and knowledge gaps. Endoscopy 2022;54(1): 52–61.
175. Deliwala SS, Mohan BP, Yarra P, et al. Efficacy & safety of EUS-directed transgastric endoscopic retrograde cholangiopancreatography (EDGE) in Roux-en-Y gastric bypass anatomy: A systematic review & meta-analysis. Surg Endosc 2023;37(6):4144–58.
176. Fusaroli P, Eloubeidi M, Calvanese C, et al. Quality of reporting in endoscopic ultrasound: Results of an international multicenter survey (the QUOREUS study). Endosc Int Open 2021;9(7):E1171–7.
177. Facciorusso A, Buccino RV, Muscatiello N. How to measure quality in endoscopic ultrasound. Ann Transl Med 2018;6(13):266.
178. Schwab R, Pahk E, Lachter J. Impact of endoscopic ultrasound quality assessment on improving endoscopic ultrasound reports and procedures. World J Gastrointest Endosc 2016;8:362–7.
179. Lachter J, Bluen B, Waxman I, et al. Establishing a quality indicator format for endoscopic ultrasound. World J Gastrointest Endosc 2013; 5(11):574–80.
180. Bruce J, Russell EM, Mollison J, et al. The quality of measurement of surgical wound infection as the basis for monitoring: A systematic review. J Hosp Infect 2001;49(2):99–108.
181. Dreisler E, Schou L, Adamsen S. Completeness and accuracy of voluntary reporting to a national case registry of laparoscopic cholecystectomy. Int J Qual Health Care 2001;13(1):51–5.
182. Ben-Menachem T, Decker GA, Early DS, et al. Adverse events of upper GI endoscopy. Gastrointest Endosc 2012;76(4):707–18.
183. Marchetti G, Ricardo VD, Ardengh AO, et al. Adverse events and mortality: Comparative analysis between diagnostic and interventional endoscopic ultrasound. Scand J Gastroenterol 2020;55(8):995–1001.
184. Eloubeidi MA, Tamhane A, Lopes TL, et al. Cervical esophageal perforations at the time of endoscopic ultrasound: A prospective evaluation of frequency, outcomes, and patient management. Am J Gastroenterol 2009;104(1):53–6.
185. Khan U, Abunassar M, Chatterjee A, et al. Advanced endoscopy trainee involvement early in EUS training may be associated with an increased risk of adverse events. J Can Assoc Gastroenterol 2020;3(2): 83–90.
186. Das A, Sivak MV Jr, Chak A. Cervical esophageal perforation during EUS: A national survey. Gastrointest Endosc 2001;53(6):599–602.
187. Mortensen MB, Fristrup C, Holm FS, et al. Prospective evaluation of patient tolerability, satisfaction with patient information, and complications in endoscopic ultrasonography. Endoscopy 2005;37(2): 146–53.
188. Razik R, James PD, Khan R, et al. Risk of adverse events associated with upper and lower endoscopic ultrasound: A population-based cohort study. Endosc Int Open 2021;9:E1427–34.
189. Wang KX, Ben QW, Jin ZD, et al. Assessment of morbidity and mortality associated with EUS-guided FNA: A systematic review. Gastrointest Endosc 2011;73(2):283–90.
190. Zhu H, Jiang F, Zhu J, et al. Assessment of morbidity and mortality associated with endoscopic ultrasound-guided fine-needle aspiration for pancreatic cystic lesions: A systematic review and meta-analysis. Dig Endosc 2017;29(6):667–75.
191. Razpotnik M, Bota S, Kutilek M, et al. The bleeding risk after endoscopic ultrasound-guided puncture of pancreatic masses. Scand J Gastroenterol 2021;56(2):205–10.
192. Barawi M, Gottlieb K, Cunha B, et al. A prospective evaluation of the incidence of bacteremia associated with EUS-guided fine-needle aspiration. Gastrointest Endosc 2001;53(2):189–92.
193. Levy MJ, Norton ID, Wiersema MJ, et al. Prospective risk assessment of bacteremia and other infectious complications in patients undergoing EUS-guided FNA. Gastrointest Endosc 2003;57(6):672–8.
194. Janssen J, König K, Knop-Hammad V, et al. Frequency of bacteremia after linear EUS of the upper GI tract with and without FNA. Gastrointest Endosc 2004;59(3):339–44.
195. Levy MJ, Norton ID, Clain JE, et al. Prospective study of bacteremia and complications with EUS FNA of rectal and perirectal lesions. Clin Gastroenterol Hepatol 2007;5(6):684–9.
196. Cho SH, Song TJ, Seo DW, et al. Efficacy and safety of EUS-guided through-the-needle microforceps biopsy sampling in categorizing the type of pancreatic cystic lesions. Gastrointest Endosc 2022;95(2): 299–309.
197. Yang D, Samarasena JB, Jamil LH, et al. Endoscopic ultrasound-guided through-the-needle microforceps biopsy in the evaluation of pancreatic cystic lesions: A multicenter study. Endosc Int Open 2018;6(12): E1423–30.
198. Karaisz FG, Elkelany OO, Davies B, et al. A review on endoscopic ultrasound-guided radiofrequency ablation (EUS-RFA) of pancreatic lesions. Diagnostics 2023;13(3):536.
199. Gollapudi LA, Tyberg A. EUS-RFA of the pancreas: Where are we and future directions. Transl Gastroenterol Hepatol 2022;7:18.
200. Kerdsirichairat T, Shin EJ. Role of endoscopic ultrasonography guided fiducial marker placement in gastrointestinal cancer. Curr Opin Gastroenterol 2020;36(5):402–8.
201. Facciorusso A, Crinò SF, Gkolfakis P, et al. Needle tract seeding after endoscopic ultrasound tissue acquisition of pancreatic lesions: A systematic review and meta-analysis. Diagnostics 2022;12(9):2113.
202. Kitano M, Yoshida M, Ashida R, et al. Needle tract seeding after endoscopic ultrasound-guided tissue acquisition of pancreatic tumors: A nationwide survey in Japan. Dig Endosc 2022;34(7):1442-55.
203. Minaga K, Takenaka M, Katanuma A, et al. Needle tract seeding: An overlooked rare complication of endoscopic ultrasound-guided fine-needle aspiration. Oncology 2017;93(Suppl 1):107–12.
204. El Chafic AH, Dewitt J, Leblanc JK, et al. Impact of preoperative endoscopic ultrasound-guided fine needle aspiration on postoperative recurrence and survival in cholangiocarcinoma patients. Endoscopy 2013;45(11):883–9.
205. Heimbach JK, Sanchez W, Rosen CB, et al. Trans-peritoneal fine needle aspiration biopsy of hilar cholangiocarcinoma is associated with disease dissemination. HPB 2011;13(5):356–60.
206. Alvarez-Sánchez MV, Jenssen C, Faiss S, et al. Interventional endoscopic ultrasonography: An overview of safety and complications. Surg Endosc 2014;28(3):712–34.
207. O'Toole TM, Schmulewitz N. Complication rates of EUS-guided celiac plexus blockade and neurolysis: Results of a large case series. Endoscopy 2009;41(7):593–7.
208. Muscatiello N, Panella C, Pietrini L, et al. Complication of endoscopic ultrasound-guided celiac plexus neurolysis. Endoscopy 2006;38(8):858.
209. Gimeno-García AZ, Elwassief A, Paquin SC, et al. Fatal complication after endoscopic ultrasound-guided celiac plexus neurolysis. Endoscopy 2012;44(Suppl 2 UCTN):E267.
210. Mittal MK, Rabinstein AA, Wijdicks EF. Pearls & oysters: Acute spinal cord infarction following endoscopic ultrasound-guided celiac plexus neurolysis. Neurology 2012;78(9):e57–9.
211. Fujii L, Clain JE, Morris JM, et al. Anterior spinal cord infarction with permanent paralysis following endoscopic ultrasound celiac plexus neurolysis. Endoscopy 2012;44(Suppl 2 UCTN):E265–6.
212. Sharaiha RZ, Khan MA, Kamal F, et al. Efficacy and safety of EUS-guided biliary drainage in comparison with percutaneous biliary drainage when ERCP fails: A systematic review and meta-analysis. Gastrointest Endosc 2017;85(5):904–14.
213. Irani S, Yang J, Khashab MA. Mitigating lumen-apposing metal stent dislodgment and allowing safe, single-stage EUS-directed transgastric ERCP. VideoGIE 2018;3(10):322–4.
214. Wang J, Hu JL, Sun SY. Endoscopic ultrasound guided gastroenterostomy: Technical details updates, clinical outcomes, and adverse events. World J Gastrointest Endosc 2023;15(11):634–40.
215. Garcia-Alonso FJ, Sanchez-Ocana R, Peñas-Herrero I, et al. Cumulative risks of stent migration and gastrointestinal bleeding in patients with lumen-apposing metal stents. Endoscopy 2018;50(4):386–95.
216. Dhir V, Bhandari S, Bapat M, et al. Comparison of EUS-guided rendezvous and precut papillotomy techniques for biliary access (with videos). Gastrointest Endosc 2012;75(2):354–9.
217. Poincloux L, Rouquette O, Buc E, et al. Endoscopic ultrasound-guided biliary drainage after failed ERCP: Cumulative experience of 101 procedures at a single center. Endoscopy 2015;47(9):794–801.
218. Tyberg A, Desai AP, Kumta NA, et al. EUS-guided biliary drainage after failed ERCP: A novel algorithm individualized based on patient anatomy. Gastrointest Endosc 2016;84(6):941–6.

219. Vila JJ, Pérez-Miranda M, Vazquez-Sequeiros E, et al. Initial experience with EUS-guided cholangiopancreatography for biliary and pancreatic duct drainage: A Spanish national survey. Gastrointest Endosc 2012;76(6):1133–41.
220. Dhindsa BS, Mashiana HS, Dhaliwal A, et al. EUS-guided biliary drainage: A systematic review and meta-analysis. Endosc Ultrasound 2020;9(2):101–9.
221. Siddiqui UD, Levy MJ. EUS-guided transluminal interventions. Gastroenterology 2018;154(7):1911–24.
222. Wang K, Zhu J, Xing L, et al. Assessment of efficacy and safety of EUS-guided biliary drainage: A systematic review. Gastrointest Endosc 2016; 83(6):1218–27.
223. Ghandour B, Bejjani M, Irani SS, et al. Classification, outcomes, and management of misdeployed stents during EUS-guided gastroenterostomy. Gastrointest Endosc 2022;95(1):80–9.
224. Tyberg A, Jha K, Shah S, et al. EUS-guided gallbladder drainage: A learning curve modified by technical progress. Endosc Int Open 2020; 8(1):E92–6.
225. Bang JY, Hawes RH, Varadarajulu S. Lumen-apposing metal stent placement for drainage of pancreatic fluid collections: Predictors of adverse events. Gut 2020;69(8):1379–81.
226. Ge PS, Young JY, Dong W, et al. EUS-guided gastroenterostomy versus enteral stent placement for palliation of malignant gastric outlet obstruction. Surg Endosc 2019;33(10):3404–11.
227. Iqbal U, Khara HS, Hu Y, et al. EUS-guided gastroenterostomy for the management of gastric outlet obstruction: A systematic review and meta-analysis. Endosc Ultrasound 2020;9(1):16–23.
228. Chen YI, Kunda R, Storm AC, et al. EUS-guided gastroenterostomy: A multicenter study comparing the direct and balloon-assisted techniques. Gastrointest Endosc 2018;87(5):1215–21.
229. Law R, Baron TH. Endoscopic ultrasound-guided gallbladder drainage. Gastrointest Endosc Clin North Am 2018;28(2):187–95.
230. Fugazza A, Sethi A, Trindade AJ, et al. International multicenter comprehensive analysis of adverse events associated with lumen-apposing metal stent placement for pancreatic fluid collection drainage. Gastrointest Endosc 2020;91(3):574–83.
231. Pawa R, Dorrell R, Nguyen M, et al. Analysis of adverse events with lumen apposing metal stents for drainage of pancreatic fluid collections. Endosc Int Open 2023;11(12):E1153–60.
232. Sharaiha RZ, Tyberg A, Khashab MA, et al. Endoscopic therapy with lumen-apposing metal stents is safe and effective for patients with pancreatic walled-off necrosis. Clin Gastroenterol Hepatol 2016;14(12): 1797–803.
233. Bukhari M, Kowalski T, Nieto J, et al. An international, multicenter, comparative trial of EUS-guided gastrogastrostomy-assisted ERCP versus enteroscopy-assisted ERCP in patients with Roux-en-Y gastric bypass anatomy. Gastrointest Endosc 2018;88(3): 486–94.
234. Krafft MR, Lorenze A, Croglio MP, et al. "Innocent as a LAMS": Does spontaneous fistula closure (secondary intention), after EUS-directed transgastric ERCP (EDGE) via 20-mm lumen-apposing metal stent, confer an increased risk of persistent fistula and unintentional weight gain? Dig Dis Sci 2022;67(6):2337–46.